



US012404847B2

(12) **United States Patent**
Schubart et al.

(10) **Patent No.: US 12,404,847 B2**

(45) **Date of Patent: *Sep. 2, 2025**

(54) **RECIPROCATING PISTON PUMP**

(56) **References Cited**

(71) Applicant: **Oshkosh Corporation**, Oshkosh, WI (US)

U.S. PATENT DOCUMENTS

(72) Inventors: **Kai P. Schubart**, Oshkosh, WI (US);
Christopher J. Rukas, Oshkosh, WI (US); **Patrick S. Dillman**, Hartford, WI (US); **Erik S. Ellifson**, Oshkosh, WI (US); **Aaron J. Rositch**, Oshkosh, WI (US)

116,572 A * 7/1871 Doyle F04B 47/04
417/404
426,476 A * 4/1890 Stitt F04B 47/04
417/404

(Continued)

OTHER PUBLICATIONS

(73) Assignee: **OSHKOSH CORPORATION**, Oshkosh, WI (US)

HydraForce Insider Blog Proper use of an Orifice in a Hydraulic Integrated Circuit Posted by David Price on Wed, Mar. 9, 2011 @ 12:03 PM Accessed from https://web.archive.org/web/20180613131529/https://info.hydraforce.com/hydraforceinsiderblog/bid/36661/Proper-u.*

(Continued)

(*) Notice: Subject to any disclaimer, the term of this patent is extended or adjusted under 35 U.S.C. 154(b) by 0 days.

This patent is subject to a terminal disclaimer.

(21) Appl. No.: **18/382,875**

Primary Examiner — Mark A Laurenzi

Assistant Examiner — Benjamin Doyle

(22) Filed: **Oct. 23, 2023**

(74) Attorney, Agent, or Firm — FOLEY & LARDNER LLP

(65) **Prior Publication Data**

US 2024/0052818 A1 Feb. 15, 2024

Related U.S. Application Data

(63) Continuation of application No. 17/961,182, filed on Oct. 6, 2022, now Pat. No. 11,815,078, which is a (Continued)

(51) **Int. Cl.**

F04B 27/067 (2006.01)

F04B 27/00 (2006.01)

F15B 15/14 (2006.01)

(52) **U.S. Cl.**

CPC **F04B 27/0673** (2013.01); **F15B 15/1447** (2013.01); **F04B 27/005** (2013.01)

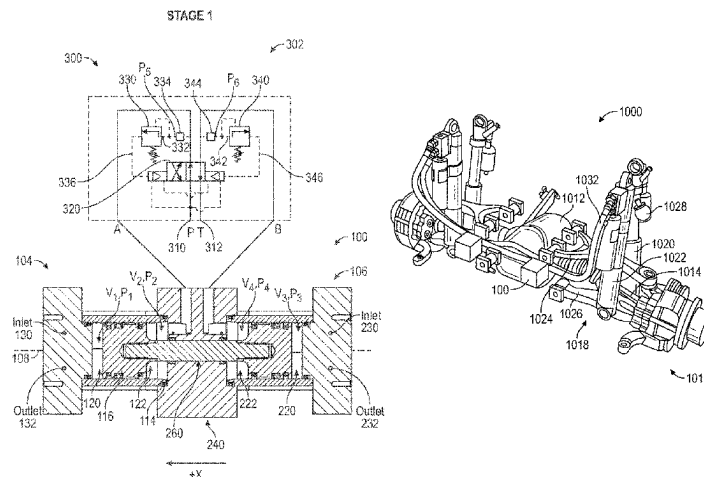
(58) **Field of Classification Search**

CPC B60G 17/0525; F04B 27/0673; F04B 27/005; F04B 27/02; F04B 27/053; (Continued)

(57) **ABSTRACT**

A pump system includes a housing defining a first internal volume and a second internal volume, a first piston positioned to separate the first internal volume into a first chamber and a second chamber, a second piston positioned to separate the second internal volume into a third chamber and a fourth chamber, a first inlet check valve configured to permit fluid flow into the first chamber, a second inlet check valve configured to permit fluid flow into the third chamber, a directional control valve (DCV) repositionable between (a) a first position where the DCV is configured to fluidly couple the second chamber to a high pressure fluid source and (b) a second position where the DCV is configured to fluidly couple the fourth chamber to the high pressure fluid source, and a relief valve configured to supply a fluid to the DCV through an orifice to move the DCV.

20 Claims, 20 Drawing Sheets



Related U.S. Application Data		4,924,671 A *	5/1990 Reinert	B30B 15/16
continuation of application No. 16/575,590, filed on Sep. 19, 2019, now Pat. No. 11,480,165.		5,013,198 A *	5/1991 Schultz	F04B 23/06
(58) Field of Classification Search		5,042,149 A *	8/1991 Holland	F04B 47/08
CPC F04B 35/008; F04B 49/22; F04B 27/08;		5,092,744 A *	3/1992 Boers	F04B 9/105
F04B 9/113; F04B 9/115; F04B 5/02;		5,145,332 A *	9/1992 Bohon	F04B 47/04
F04B 9/105; F15B 15/1447; F15B 3/00;		5,170,691 A *	12/1992 Baatrup	F04B 9/107
F15B 7/00; F15B 15/00; F01L 5/00;		5,249,502 A *	10/1993 Radocaj	F15B 15/1409
F01L 15/00; F01L 2740/00; F01L 29/00;		5,249,933 A *	10/1993 Moody	F04B 7/0076
F01L 21/00		5,324,175 A *	6/1994 Sorensen	F04B 9/1295
USPC 417/403, 404		5,337,561 A *	8/1994 Raghavan	F15B 3/00
See application file for complete search history.		5,415,531 A *	5/1995 Cavanaugh	F04B 9/133
(56) References Cited		5,462,414 A *	10/1995 Permar	F04B 9/115
U.S. PATENT DOCUMENTS		5,564,912 A *	10/1996 Peck	F04D 13/04
764,156 A * 7/1904 Spencer		5,616,005 A *	4/1997 Whitehead	F04F 1/10
1,573,584 A * 2/1926 Smith		5,639,119 A *	6/1997 Plate	B62D 49/08
1,674,614 A * 6/1928 Berkman		5,941,508 A	8/1999 Murata et al.	
3,207,442 A * 9/1965 Kessler		6,129,368 A *	10/2000 Ishikawa	F16F 9/064
3,215,339 A * 11/1965 Jackson		6,135,724 A *	10/2000 Yoder	F04B 13/00
3,249,289 A * 5/1966 Maglott		6,203,696 B1 *	3/2001 Pearson	F04B 9/115
3,374,863 A * 3/1968 Kozlowski		6,231,322 B1 *	5/2001 Adolfsson	B67D 1/004
3,540,349 A * 11/1970 Pennther		6,266,959 B1	7/2001 Markwart	
3,700,360 A * 10/1972 Shaddock		6,299,766 B1 *	10/2001 Permar	B01D 61/10
3,722,366 A * 3/1973 Richardson		6,418,970 B1	7/2002 Deul	
3,771,421 A * 11/1973 Scheffer		6,435,843 B1 *	8/2002 Hur	F04B 9/113
3,811,795 A * 5/1974 Olsen		6,478,552 B1 *	11/2002 Batten	F04B 9/135
3,838,710 A * 10/1974 Reip		6,491,813 B2 *	12/2002 Verde	B01D 61/06
4,021,156 A * 5/1977 Fuchs, Jr.		6,568,911 B1 *	5/2003 Brightwell	F04B 15/06
4,029,440 A * 6/1977 Olsen		6,761,027 B2 *	7/2004 A'Hearn	E02F 9/2225
4,062,639 A * 12/1977 Conlee		6,776,080 B2 *	8/2004 Hansen	F15B 3/00
4,220,074 A * 9/1980 Reip		7,175,395 B1 *	2/2007 Forest	F04B 9/115
4,347,049 A * 8/1982 Anderson		7,316,363 B2	1/2008 Hume et al.	
4,367,140 A * 1/1983 Wilson		7,354,252 B2 *	4/2008 Baatrup	F15B 3/00
4,368,008 A * 1/1983 Budzich		7,527,483 B1 *	5/2009 Glauber	F04B 43/0736
4,373,865 A * 2/1983 Budzich		8,037,678 B2 *	10/2011 McBride	F15B 1/024
4,383,804 A * 5/1983 Budzich		8,323,003 B2 *	12/2012 Judge	F04B 9/1376
4,386,888 A * 6/1983 Verley		8,333,390 B2	12/2012 Linsmeier et al.	
4,469,315 A 9/1984 Nicholls et al.		8,376,719 B2	2/2013 Grady et al.	
4,490,096 A * 12/1984 Box		8,465,025 B2	6/2013 Venton-Walters et al.	
4,515,516 A * 5/1985 Perrine		8,596,648 B2	12/2013 Venton-Walters et al.	
4,524,582 A * 6/1985 Lucas		8,764,029 B2	7/2014 Venton-Walters et al.	
4,653,986 A * 3/1987 Ashton		8,801,017 B2	8/2014 Ellifson et al.	
4,666,374 A * 5/1987 Nelson		8,801,393 B2	8/2014 Crabtree et al.	
		8,821,130 B2	9/2014 Venton-Walters et al.	
		8,876,133 B2	11/2014 Ellifson	
		8,991,834 B2	3/2015 Venton-Walters et al.	

(56)

References Cited**U.S. PATENT DOCUMENTS**

9,127,657 B2 *	9/2015	Glauber	F04B 9/131	2013/0249175 A1 *	9/2013	Ellifson	F16F 9/096 188/314
9,127,738 B2	9/2015	Ellifson et al.		2013/0287615 A1 *	10/2013	Fong	F04B 3/00 417/523
9,174,686 B1	11/2015	Messina et al.		2014/0378253 A1 *	12/2014	Tanaka	F16H 7/1236 474/110
9,291,230 B2	3/2016	Ellifson et al.		2015/0008420 A1	1/2015	Mazhari et al.	
9,303,715 B2	4/2016	Dillman et al.		2015/0128584 A1 *	5/2015	Zaleski	F02N 9/04 60/459
9,327,576 B2	5/2016	Ellifson		2015/0308420 A1 *	10/2015	Donnally	F04B 47/02 417/404
9,440,829 B2	9/2016	Sadiq		2015/0361970 A1 *	12/2015	White	F04B 53/143 417/320
9,581,153 B2	2/2017	Venton-Walters et al.		2016/0257390 A1 *	9/2016	Tsutsui	C23F 13/06
9,688,112 B2	6/2017	Venton-Walters et al.		2017/0321728 A1 *	11/2017	Iversen	F04F 13/00
9,707,869 B1	7/2017	Messina et al.		2018/0149145 A1 *	5/2018	Sato	F04B 9/113
9,765,841 B2	9/2017	Ellifson et al.		2018/0335104 A1	11/2018	Dillman et al.	
9,809,080 B2	11/2017	Ellifson et al.		2019/0120258 A1 *	4/2019	Zavadinka	F15B 11/0325
9,890,771 B2 *	2/2018	Jiang	F04B 53/10	2019/0154023 A1 *	5/2019	Oroojloo	F04B 39/08
9,944,145 B2	4/2018	Dillman et al.		2019/0178329 A1	6/2019	Dumitru et al.	
10,030,737 B2	7/2018	Dillman et al.		2019/0185077 A1	6/2019	Smith et al.	
10,350,956 B2	7/2019	Dillman et al.		2019/0262646 A1	8/2019	Linsmeier et al.	
10,369,860 B2	8/2019	Ellifson et al.		2019/0308480 A1	10/2019	Dillman et al.	
10,408,196 B2 *	9/2019	Huang	F04B 49/06	2019/0316650 A1	10/2019	Dillman et al.	
10,421,332 B2	9/2019	Venton-Walters et al.		2019/0337348 A1	11/2019	Venton-Walters et al.	
10,422,403 B2	9/2019	Ellifson et al.		2019/0337350 A1	11/2019	Ellifson et al.	
10,584,699 B2 *	3/2020	Huang	F04B 5/02	2019/0338823 A1	11/2019	Ellifson et al.	
10,920,587 B2 *	2/2021	Yuratich	E21B 49/10	2020/0040882 A1 *	2/2020	Kalmari	F16N 7/38
2003/0118459 A1 *	6/2003	Gerhardt	F04B 49/065 417/390				
2007/0045067 A1 *	3/2007	Schedgick	B60G 17/0152 188/266				
2008/0008025 A1 *	1/2008	Abe	B28C 5/422 366/61				
2009/0077957 A1 *	3/2009	Noble	F15B 15/204 417/404				
2011/0030364 A1 *	2/2011	Persson	F15B 21/08 60/486				
2011/0192157 A1 *	8/2011	Ogawa	F15B 21/14 60/473				
2011/0225961 A1 *	9/2011	McCrary	F15B 21/12 60/415				

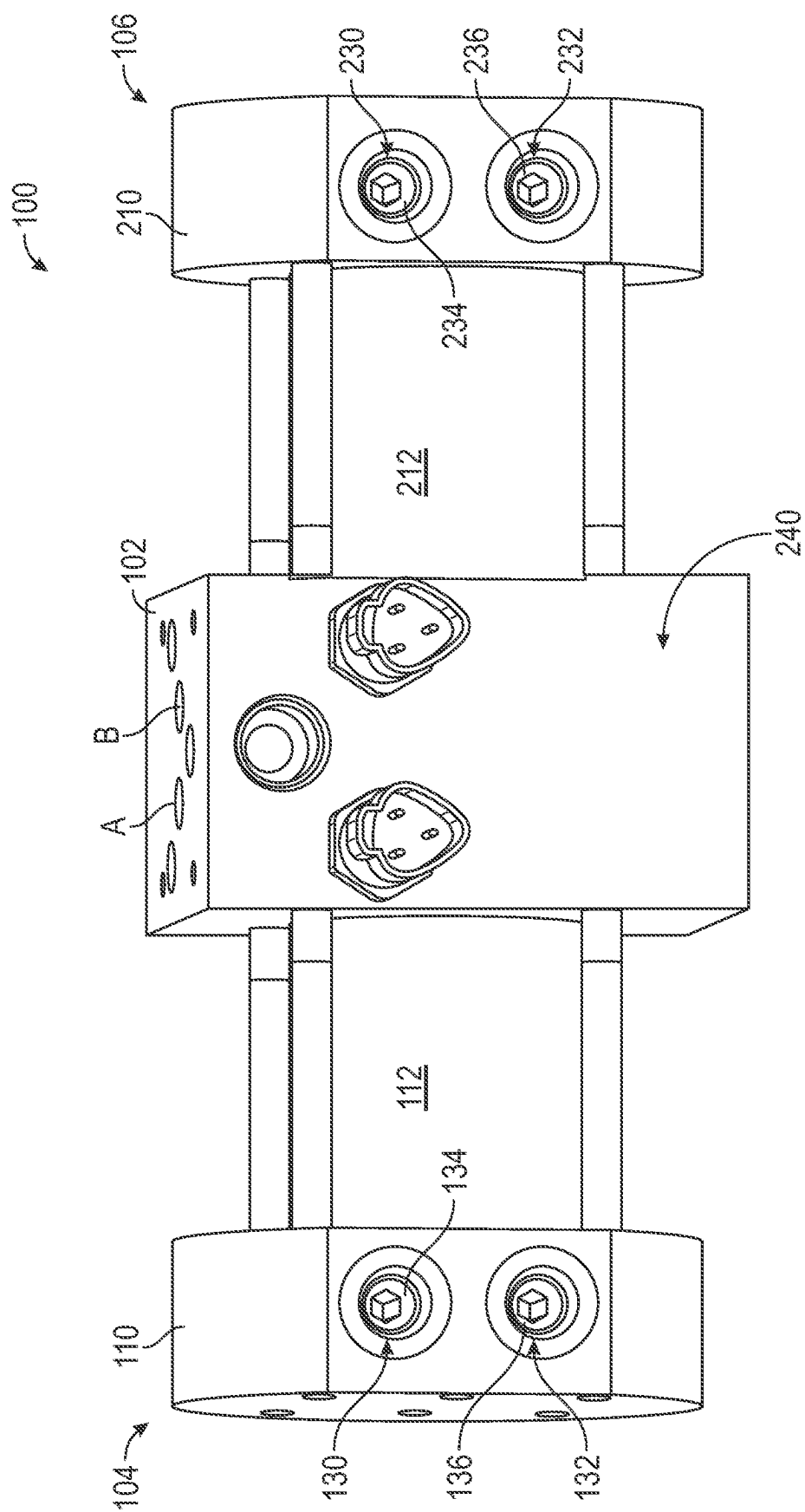
OTHER PUBLICATIONS

U.S. Appl. No. 15/956,974, filed Apr. 19, 2018, Oshkosh Defense, LLC.

U.S. Appl. No. 15/956,992, filed Apr. 19, 2018, Oshkosh Defense, LLC.

U.S. Appl. No. 16/575,590, filed Sep. 19, 2019, Schubart et al.

* cited by examiner



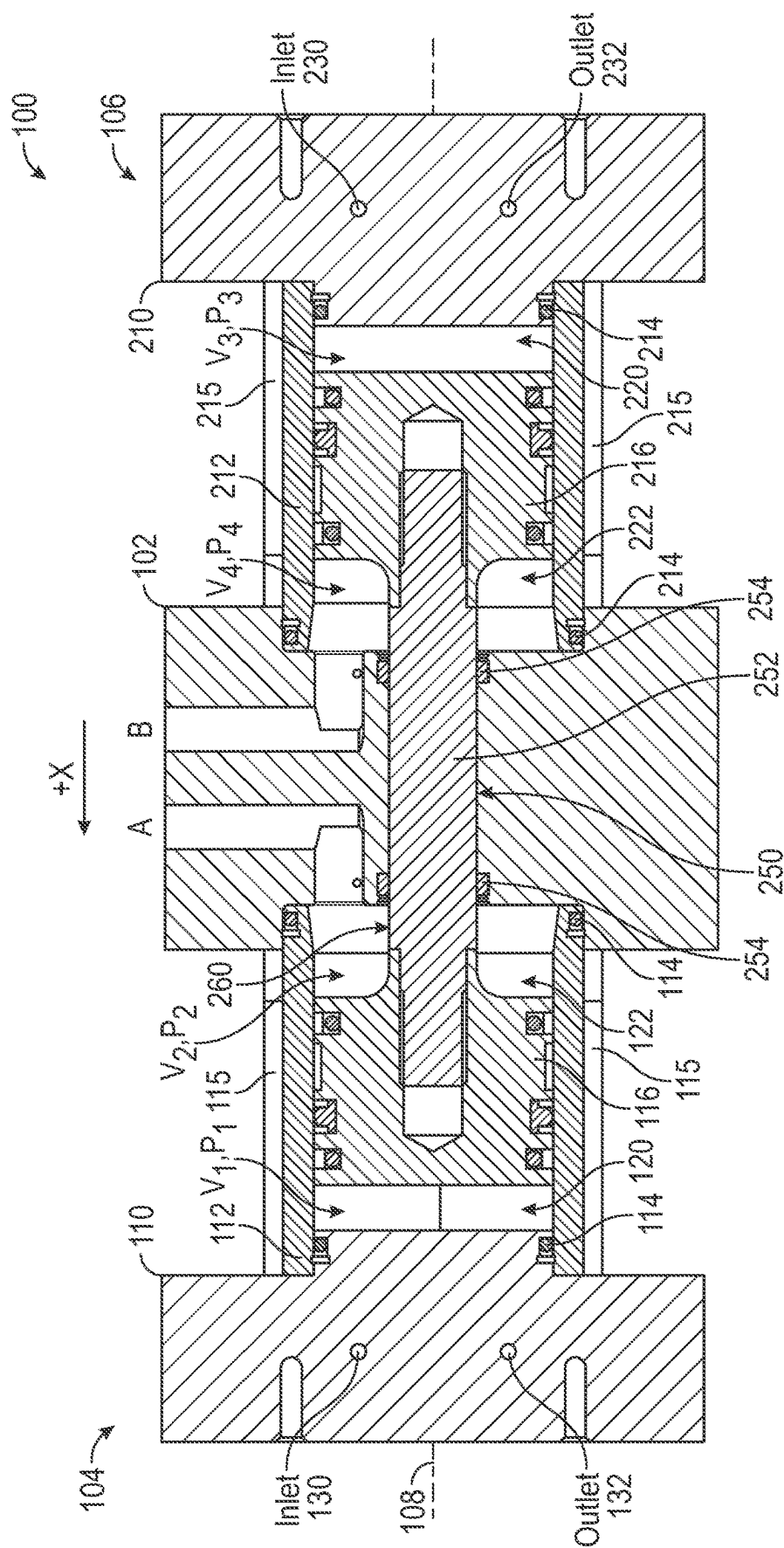


FIG. 2

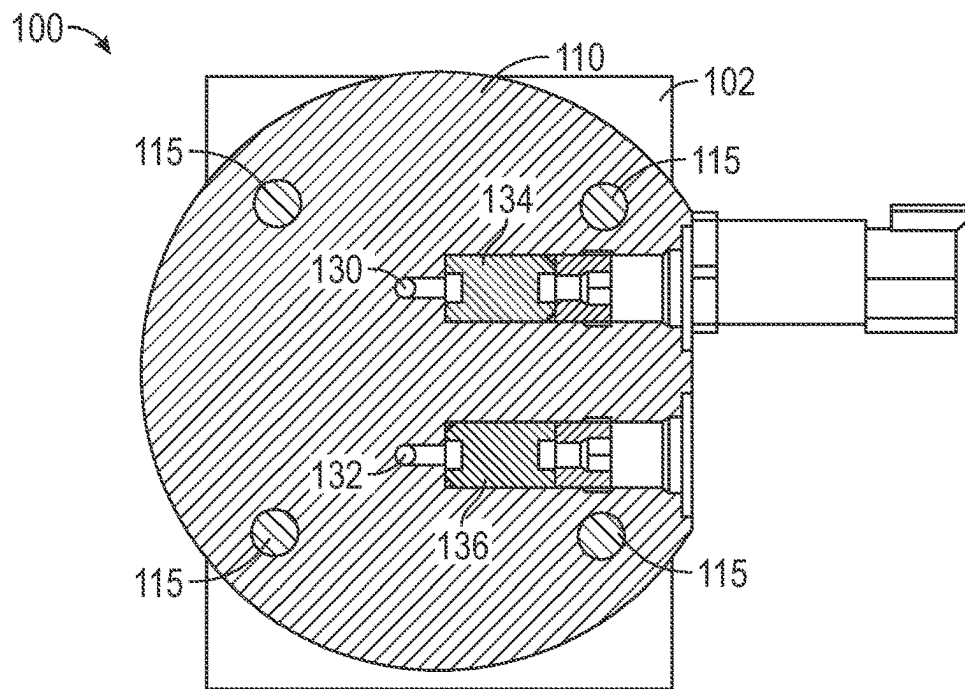


FIG. 3

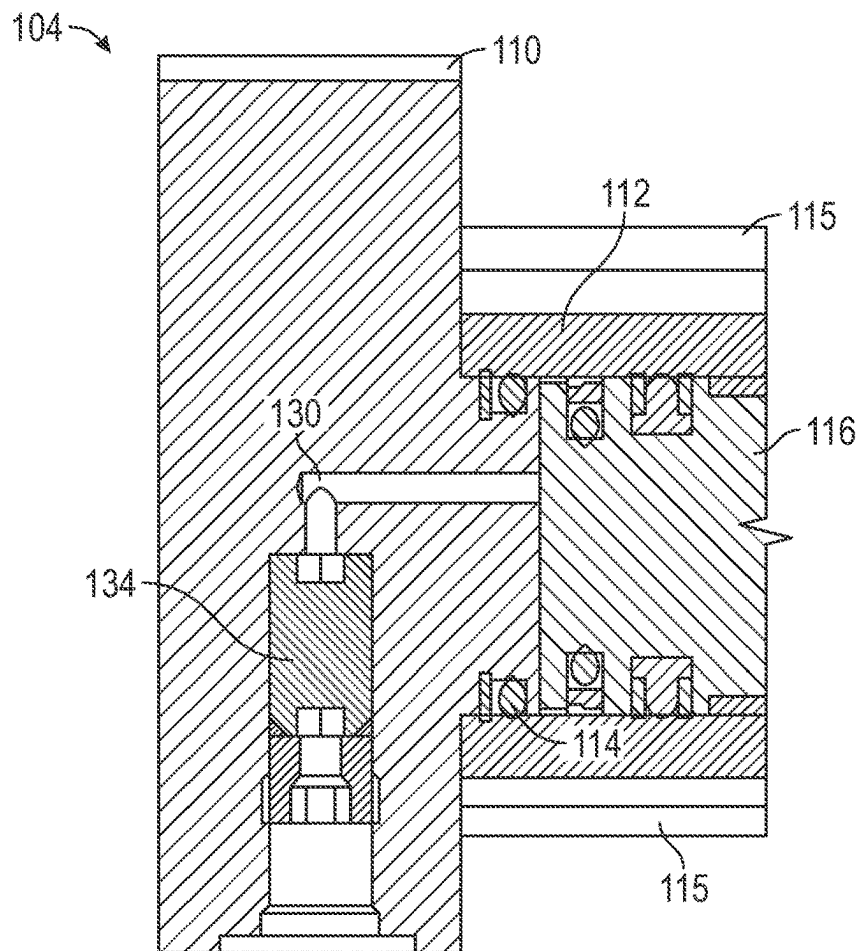


FIG. 4

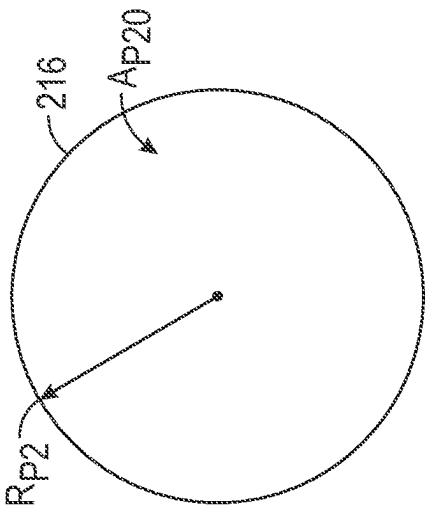


FIG. 5

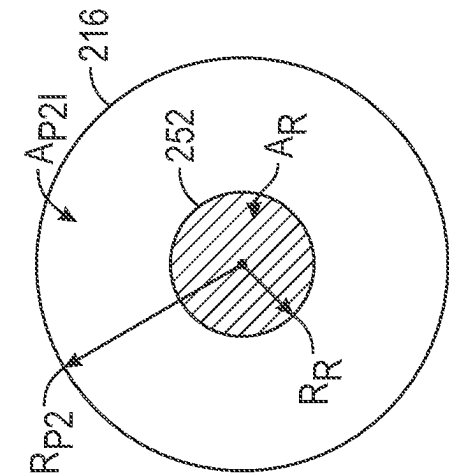


FIG. 6

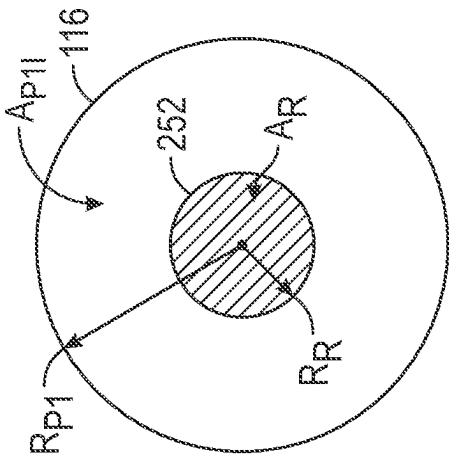


FIG. 7

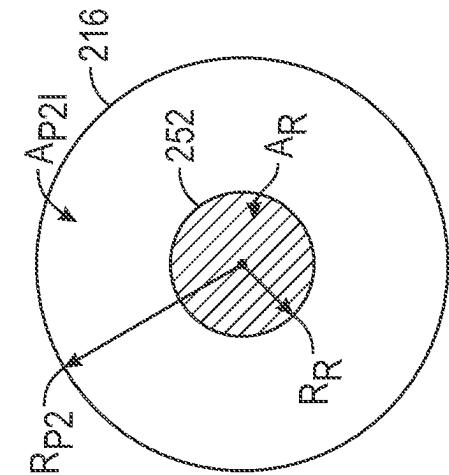


FIG. 8

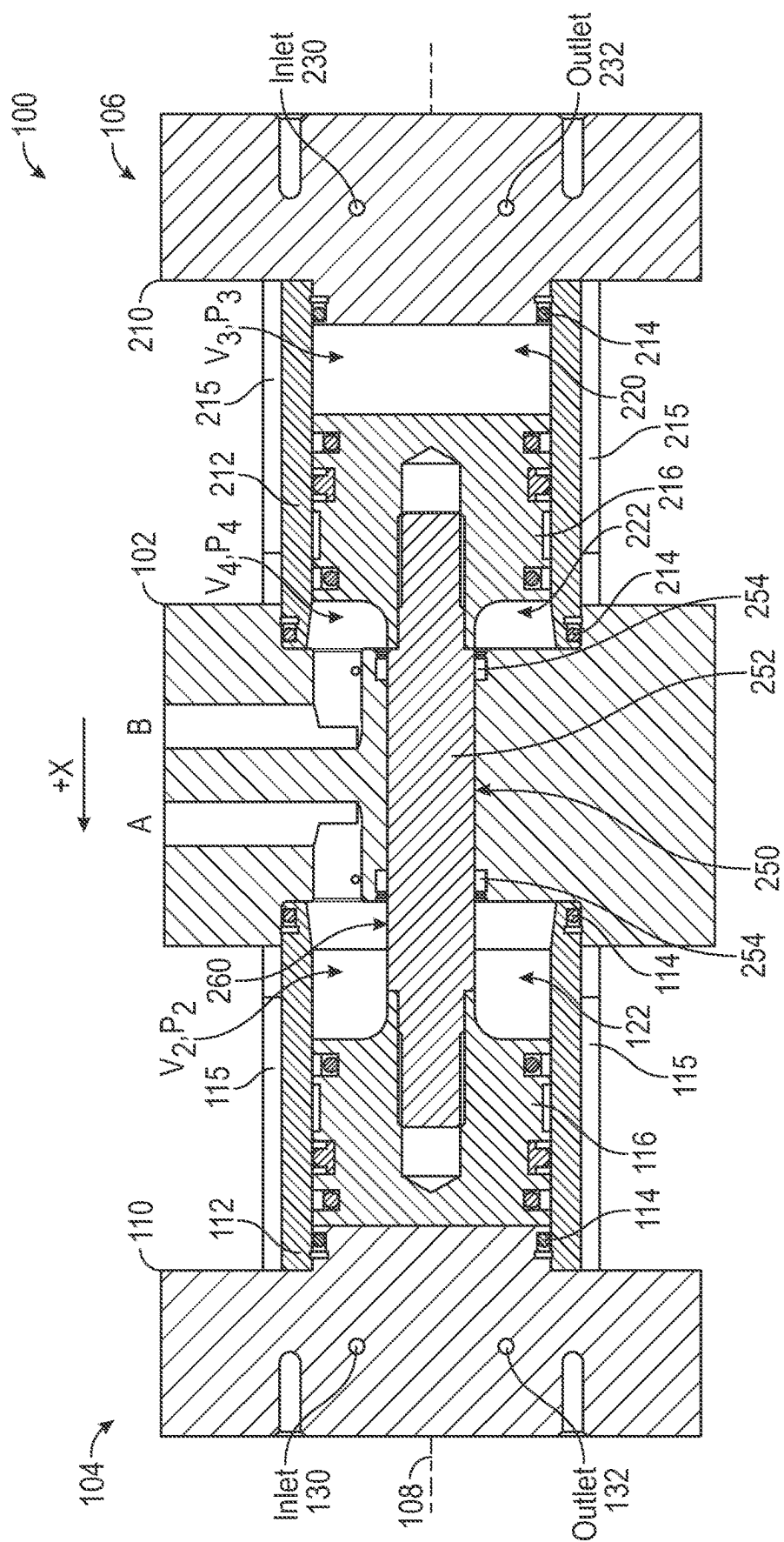


FIG. 9

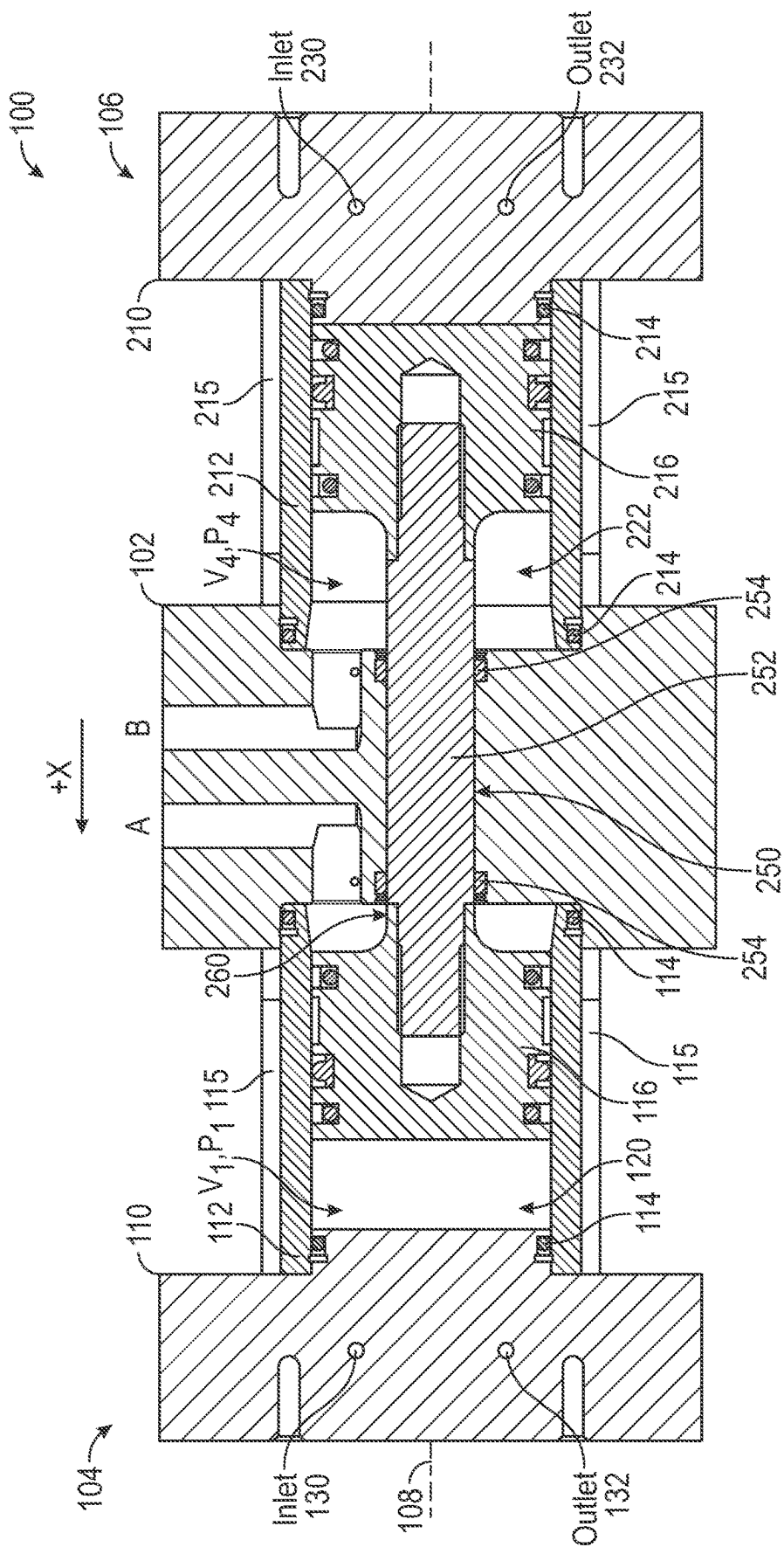


FIG. 10

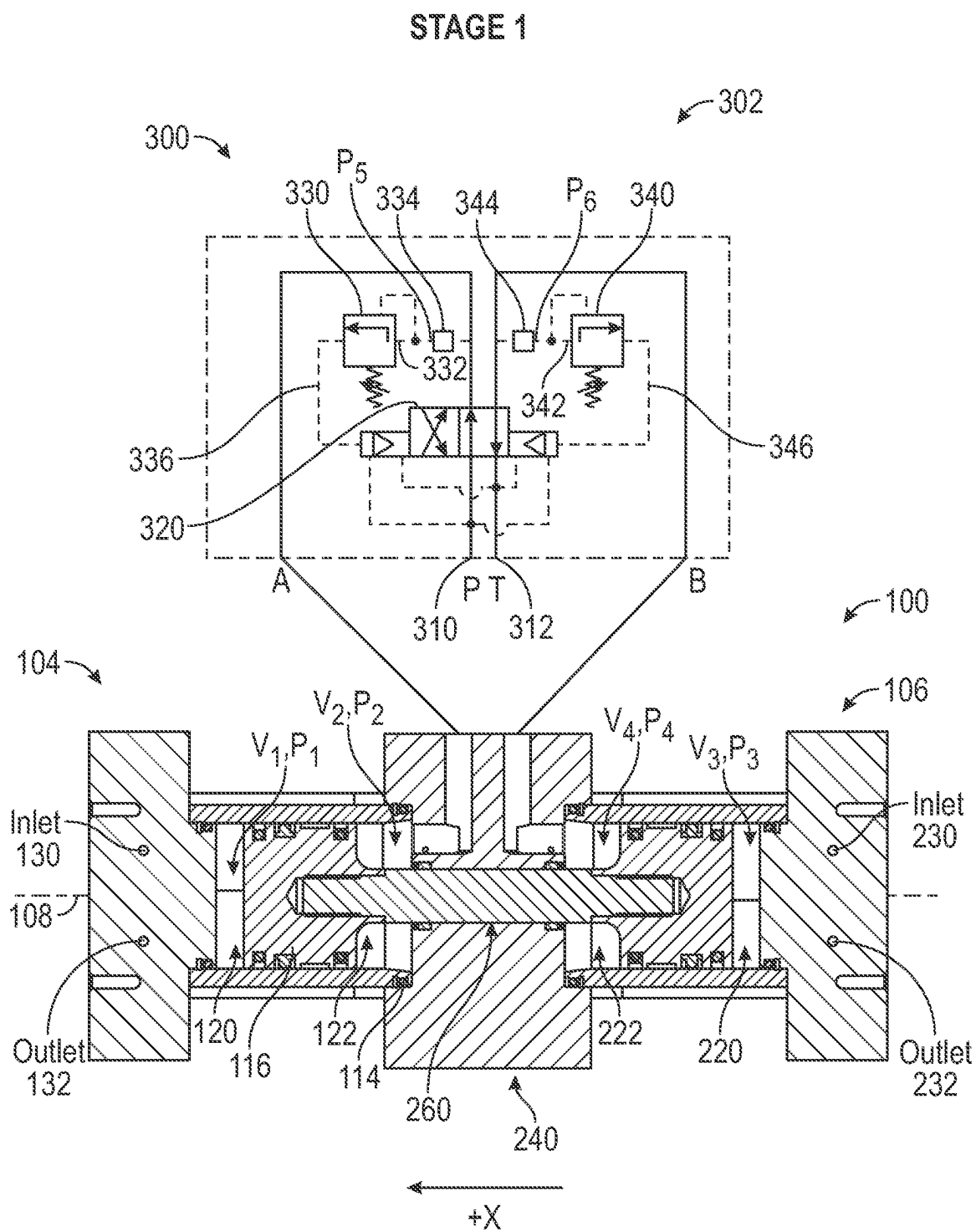


FIG. 11

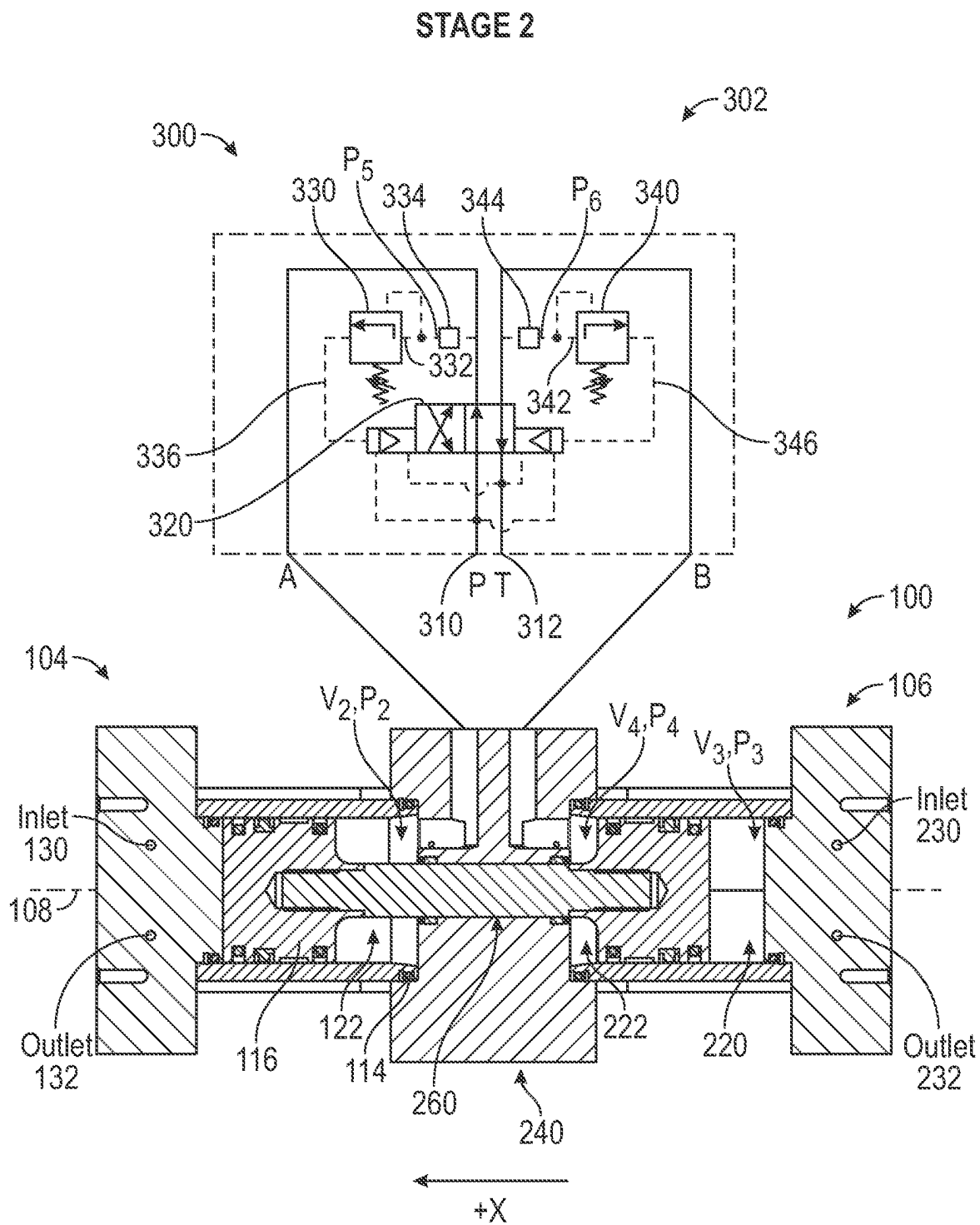


FIG. 12

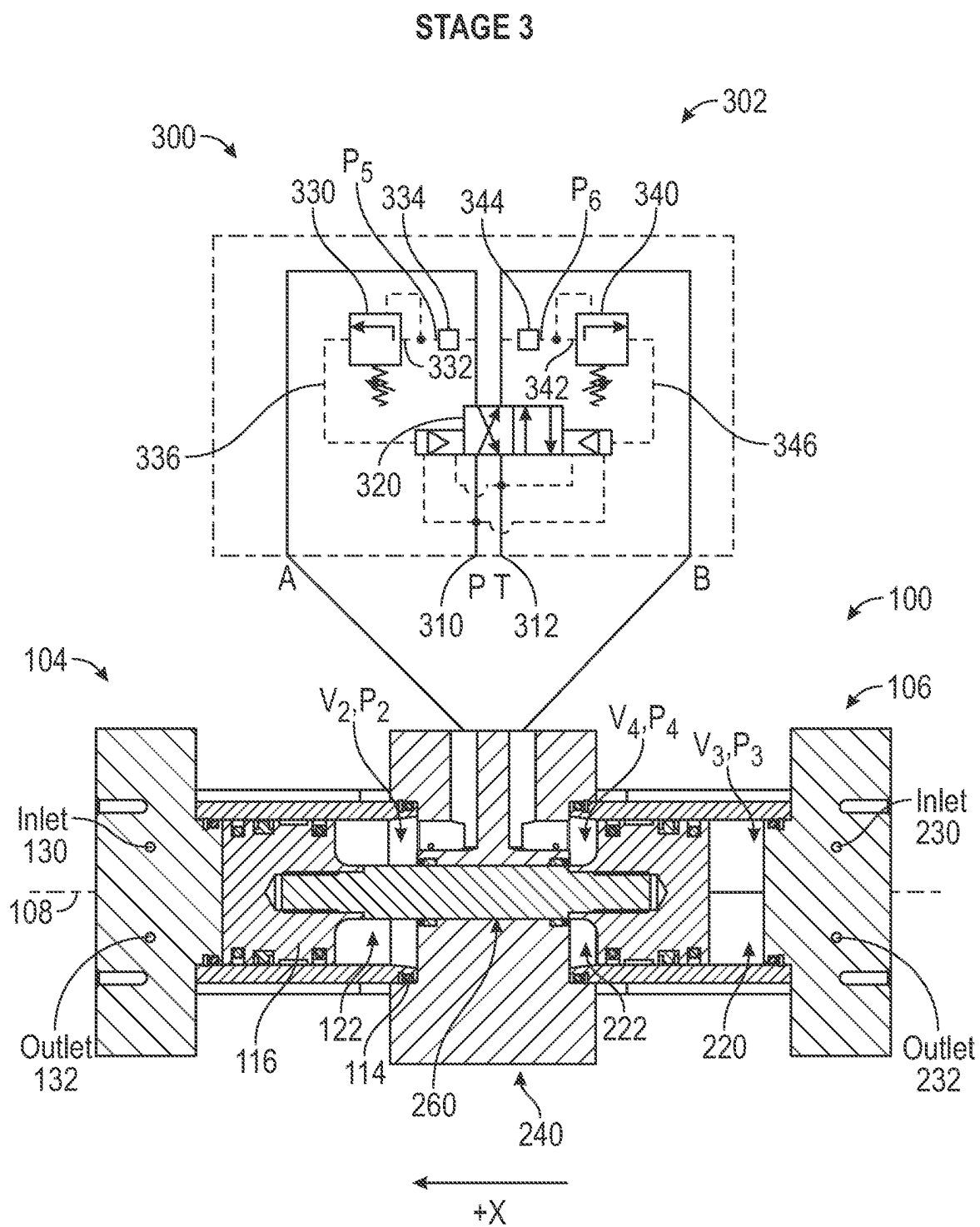


FIG. 13

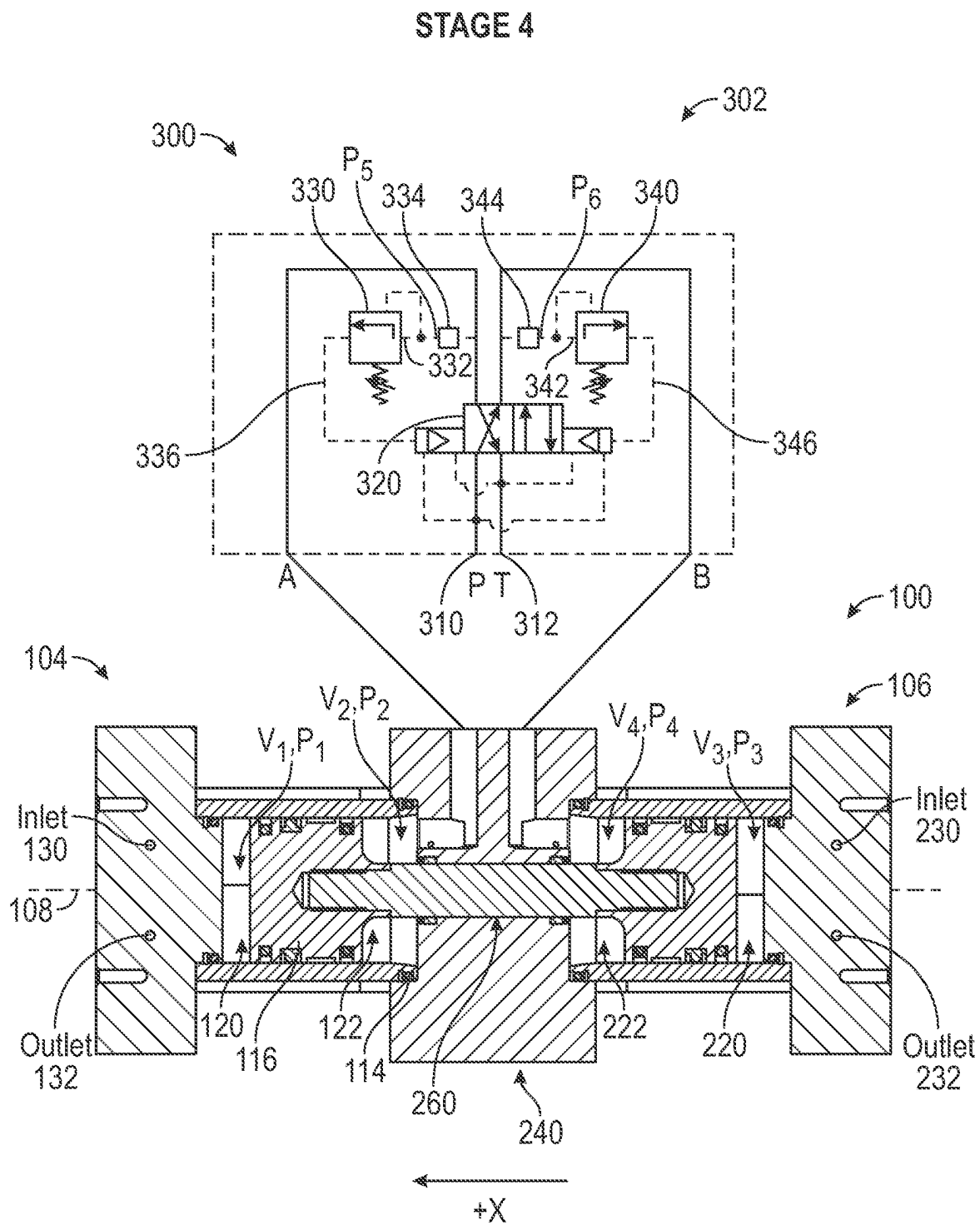


FIG. 14

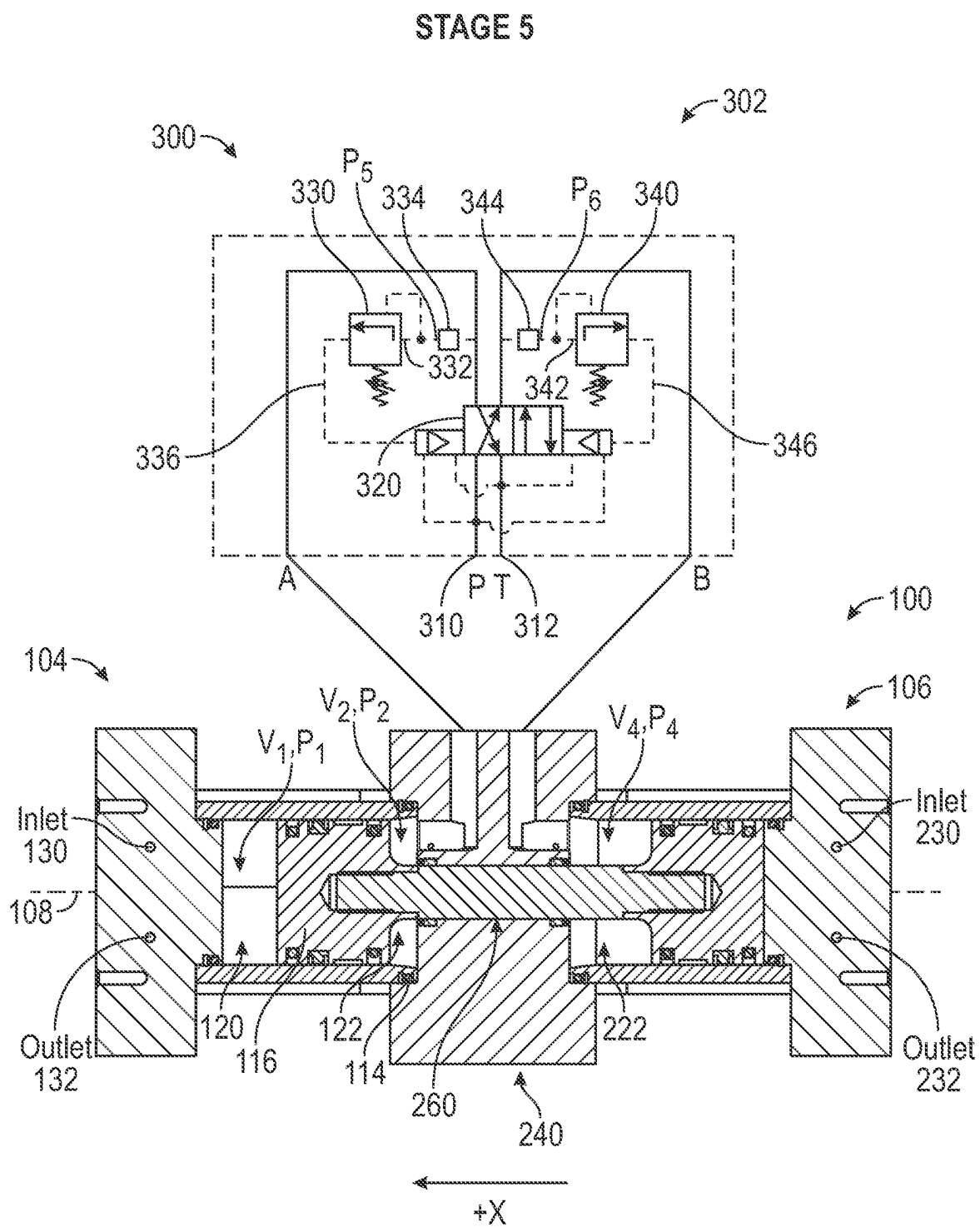


FIG. 15

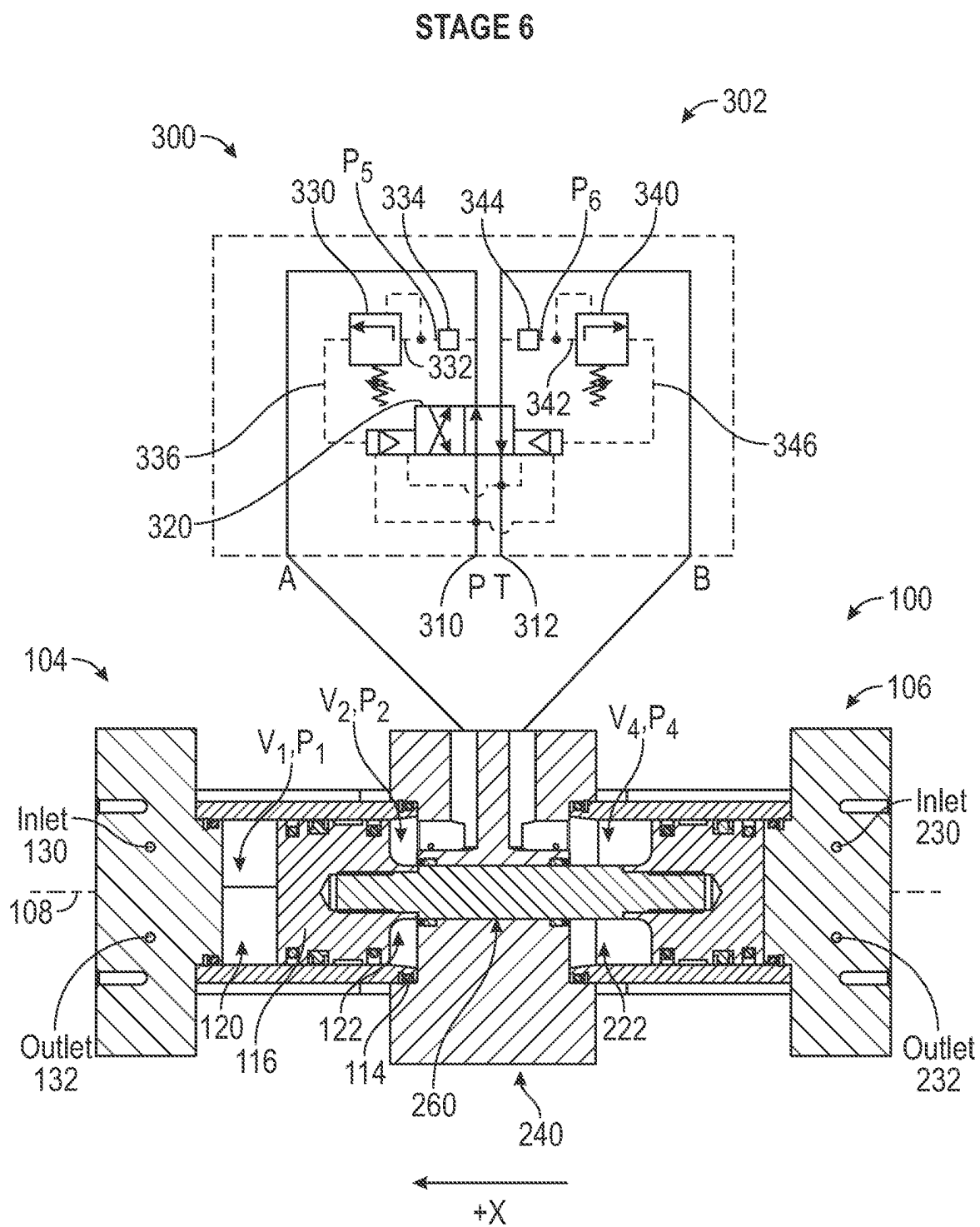


FIG. 16

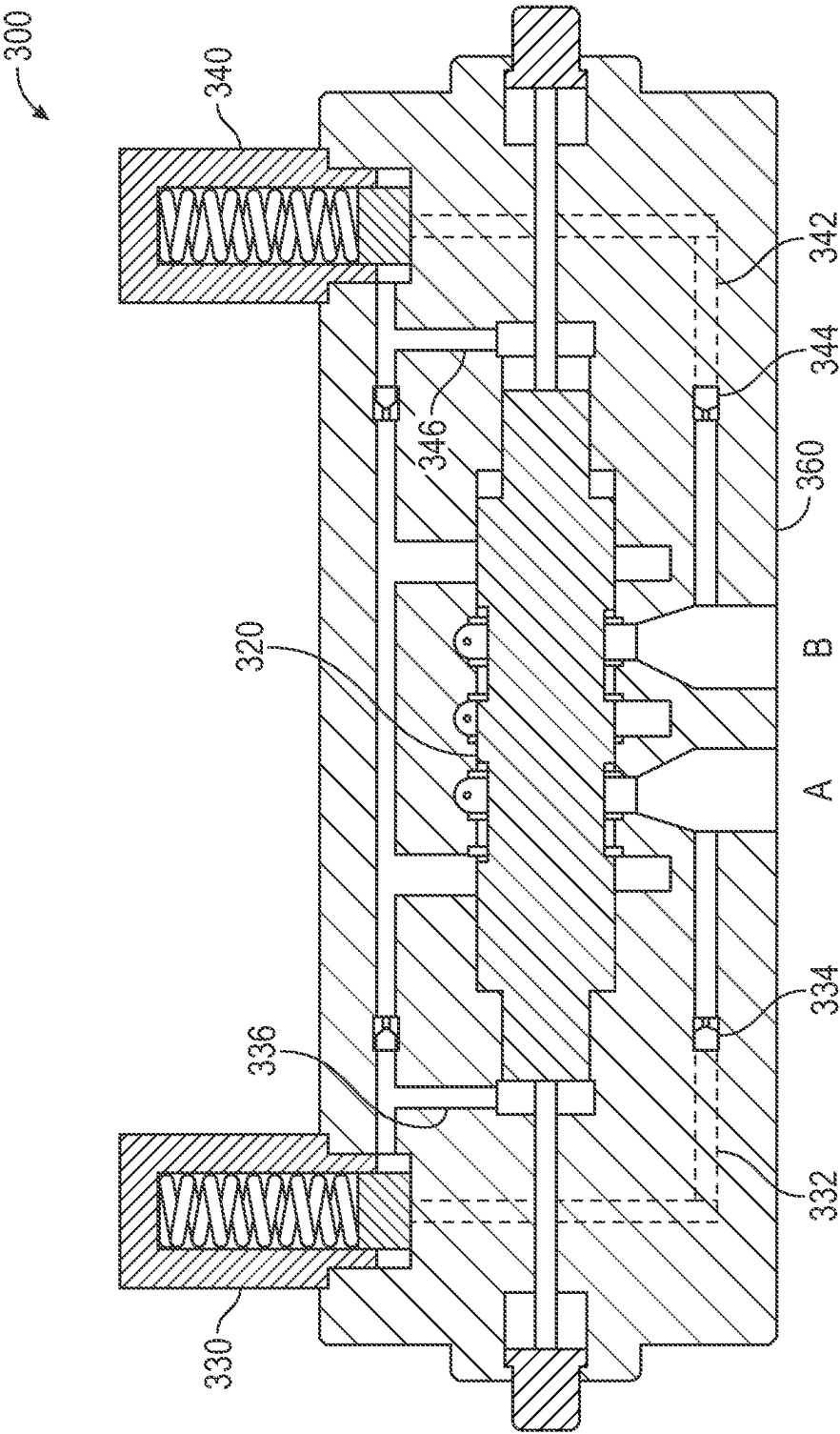
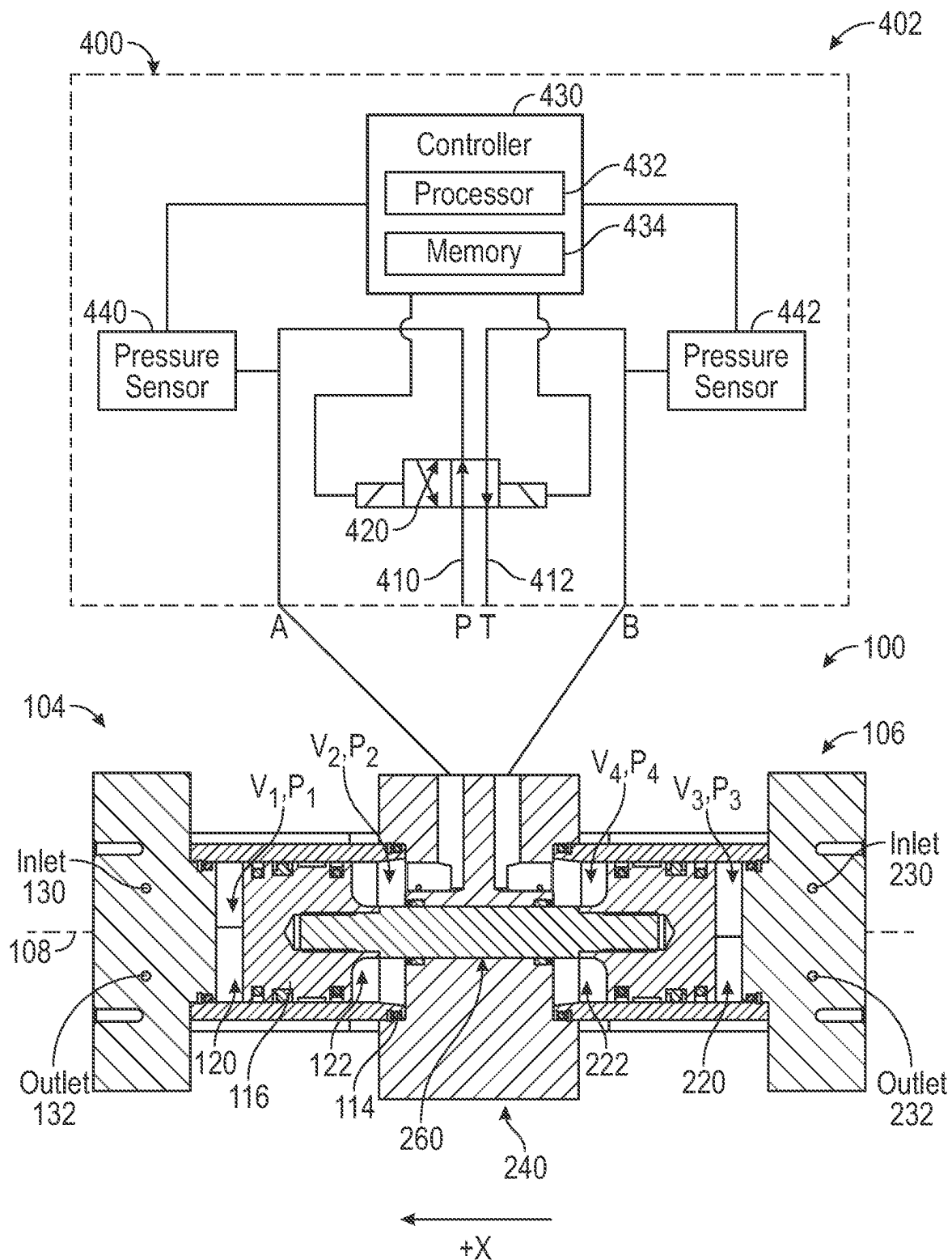


FIG. 17



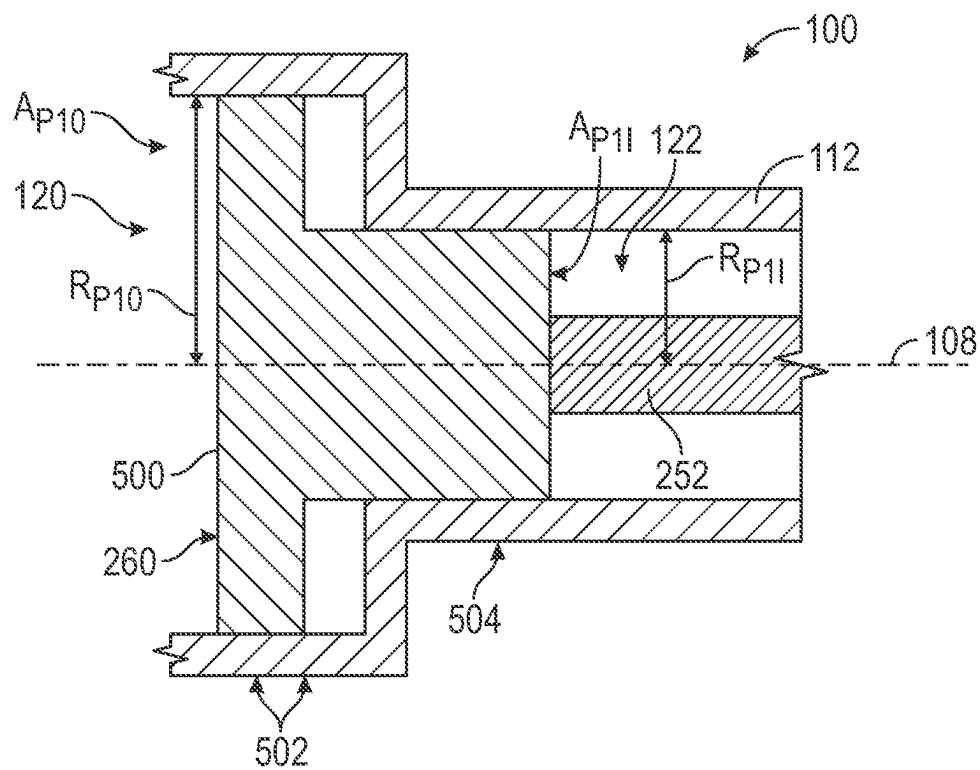


FIG. 19

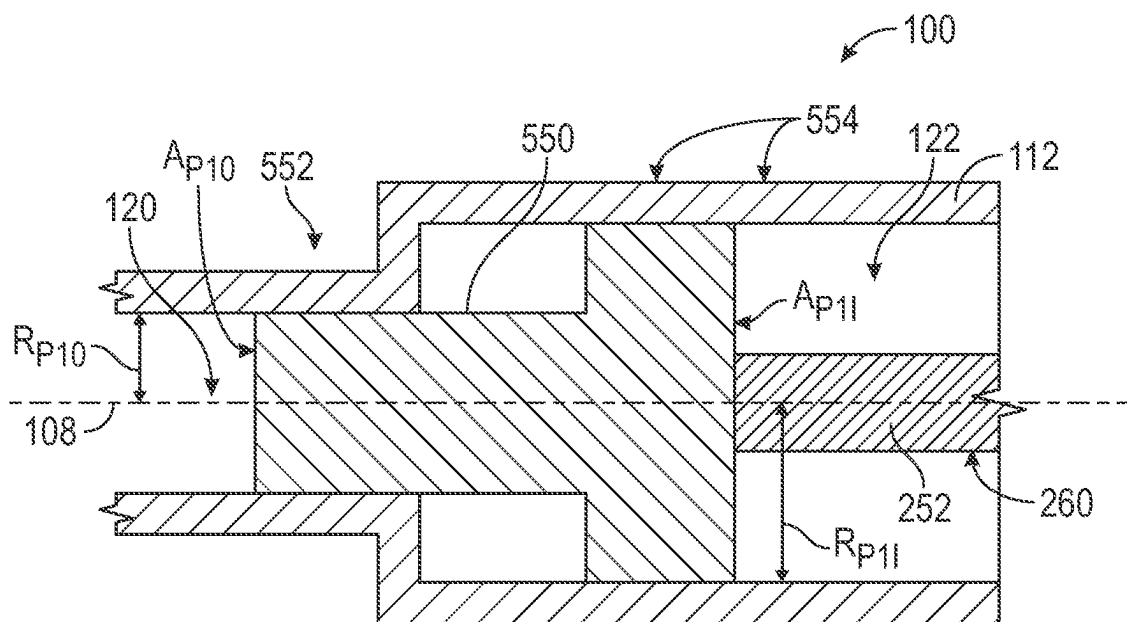


FIG. 20

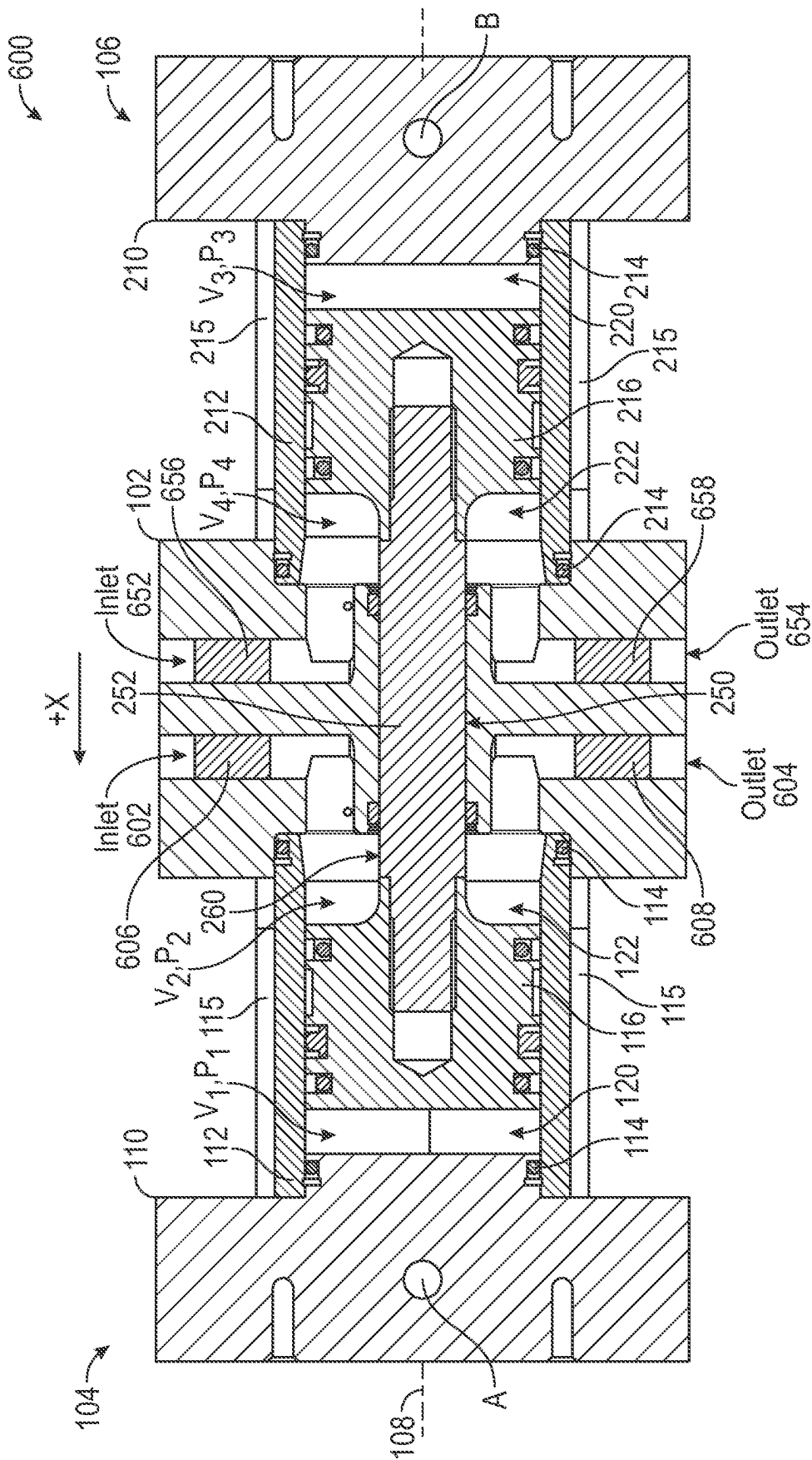


FIG. 21

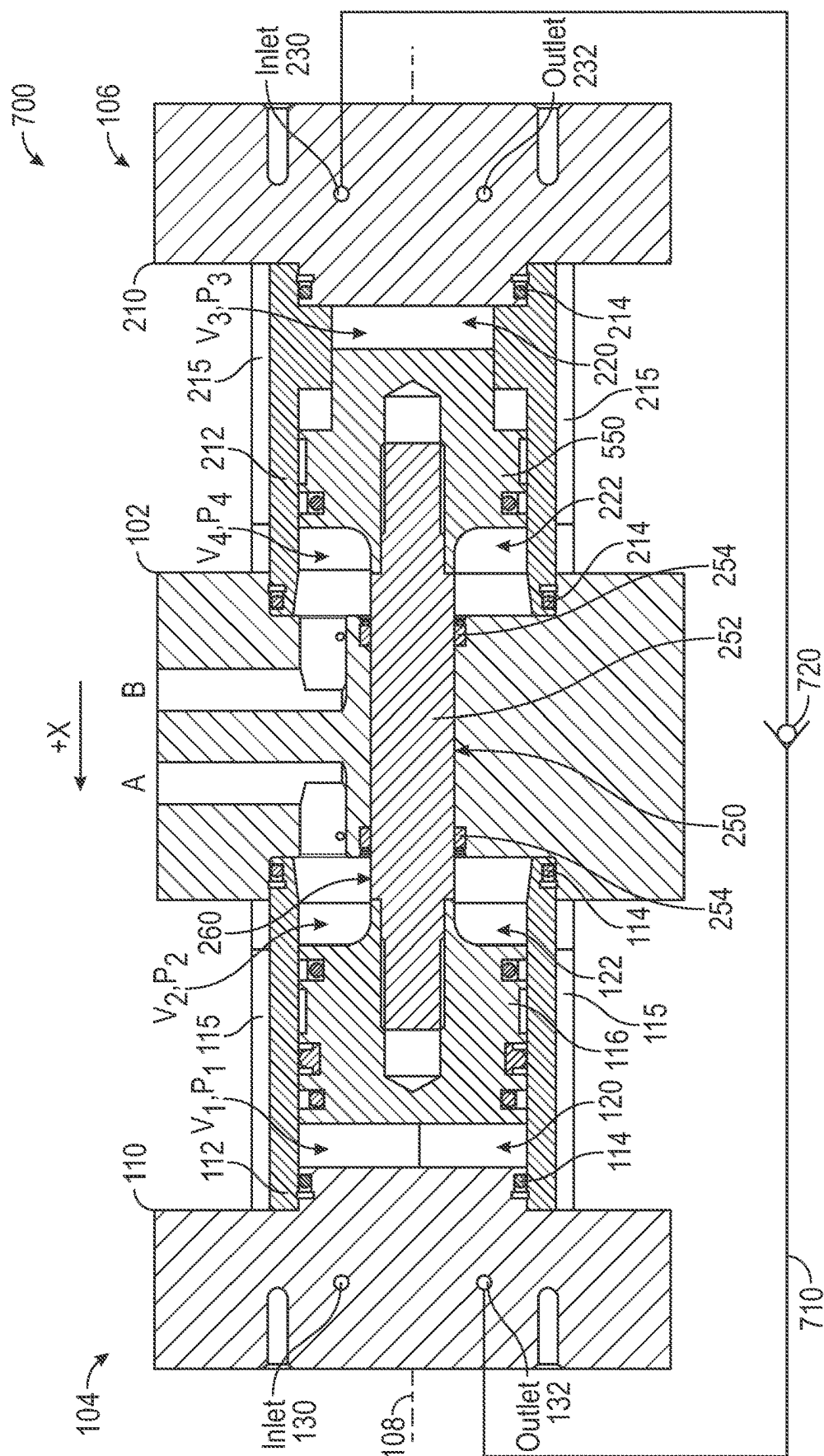


FIG. 22

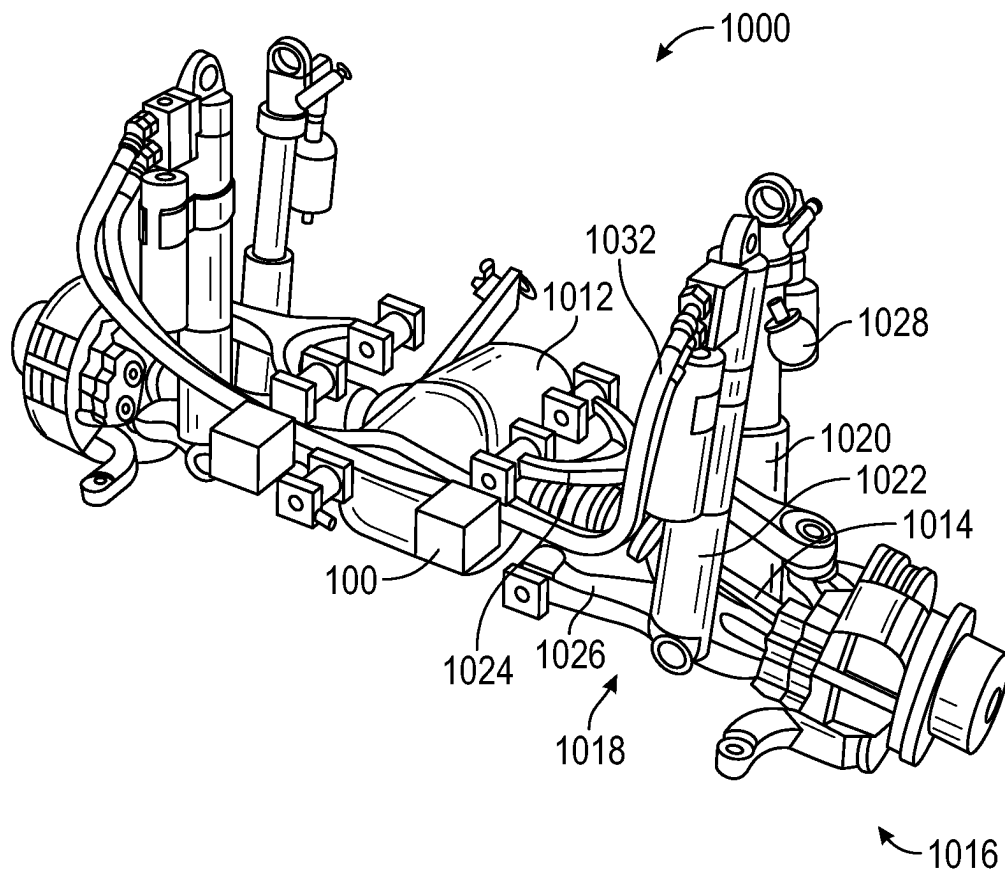


FIG. 23

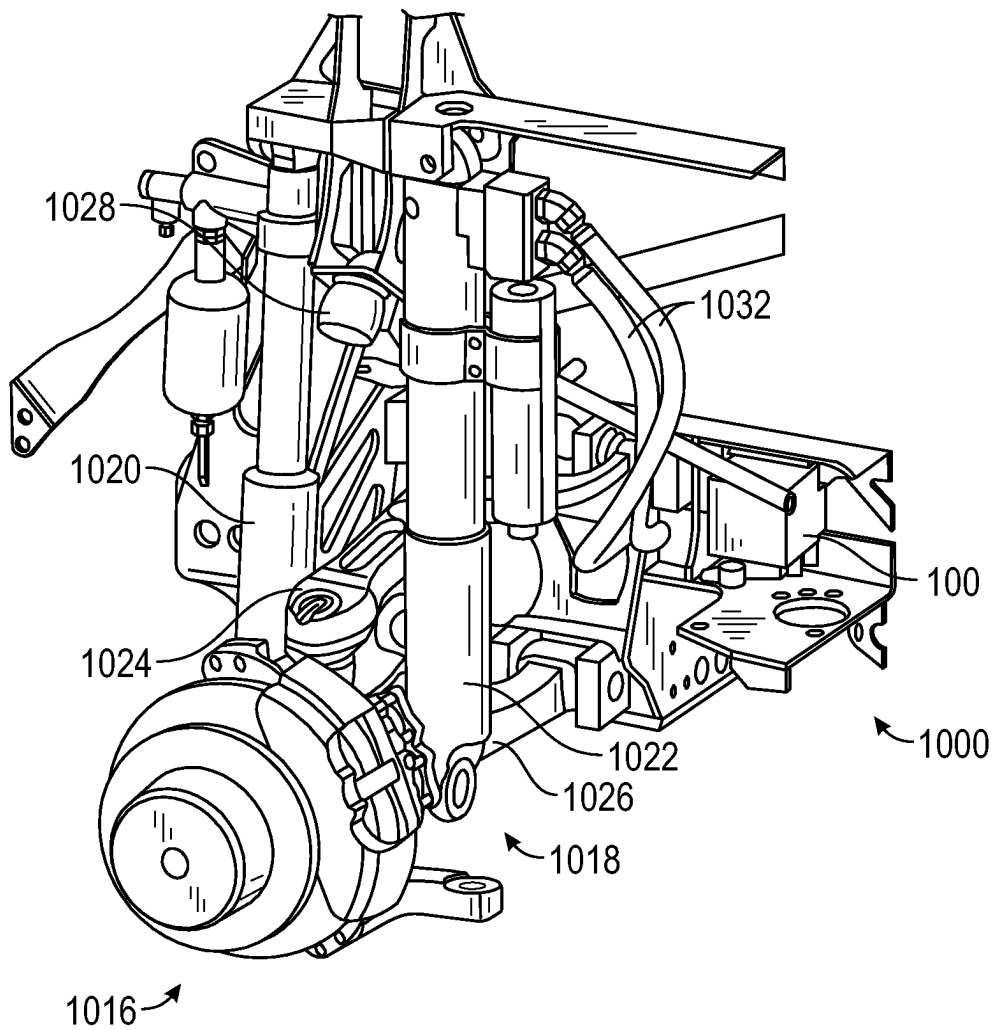


FIG. 24

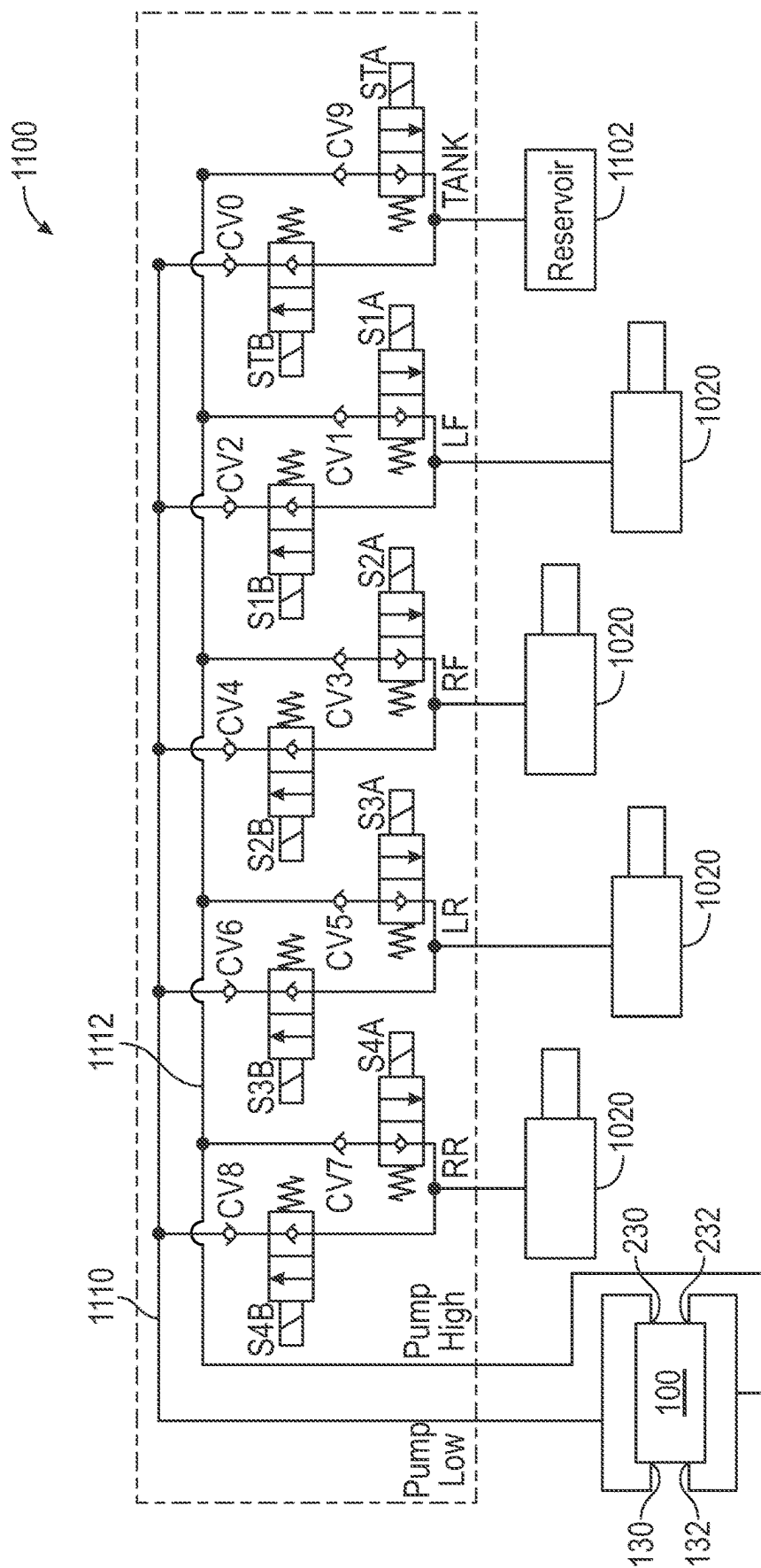


FIG. 25

1

RECIPROCATING PISTON PUMP**CROSS-REFERENCE TO RELATED PATENT APPLICATION**

This application is a continuation of U.S. patent application Ser. No. 17/961,182, filed Oct. 6, 2022, which is a continuation of U.S. patent application Ser. No. 16/575,590, filed Sep. 19, 2019, now U.S. Pat. No. 11,480,165, both of which are incorporated herein by reference in their entireties.

BACKGROUND

The present disclosure relates generally to pumps and compressors. More specifically, the present disclosure relates to a pump for conveying compressed gas throughout a vehicle.

Gas springs are used within vehicle suspension systems to connect a wheel-end assembly to a chassis of the vehicle. A length of each gas spring is controlled by a volume of pressurized gas contained within the gas spring. When a wheel encounters an obstruction (e.g., a bump, a pothole, etc.), the gas compresses or expands to permit the wheel to maintain contact with the ground while minimizing movement of the chassis. Some vehicles include pumps that vary the amount of gas within the gas spring to vary a ride height of the vehicle. It is desirable for such pumps to have as high of a flow rate as possible to facilitate rapid adjustment of ride height.

SUMMARY

At least one embodiment relates to a pump system. The pump system includes a housing, a first piston, a second piston, a first inlet check valve, a second inlet check valve, a directional control valve (DCV), and a relief valve. The housing defines a first internal volume and a second internal volume. The first piston is positioned to separate the first internal volume into a first chamber and a second chamber. The second piston is positioned to separate the second internal volume into a third chamber and a fourth chamber. The second piston is coupled to the first piston. The first inlet check valve is configured to be at least selectively fluidly coupled to a reservoir. The first inlet check valve is positioned to permit a first fluid received from the reservoir to enter the first chamber. The second inlet check valve is configured to be at least selectively fluidly coupled to the reservoir. The second inlet check valve is positioned to permit the first fluid received from the reservoir to enter the third chamber. The DCV is fluidly coupled to the second chamber and the fourth chamber. The DCV is configured to alternately supply a second fluid to the second chamber and the fourth chamber such that the first fluid is forced out of one of the first chamber or the third chamber. The DCV is repositionable between (a) a first position where the DCV is configured to fluidly couple the second chamber to a high pressure fluid source and (b) a second position where the DCV is configured to fluidly couple the fourth chamber to the high pressure fluid source. The relief valve is fluidly coupled to the second chamber through a sensing line. The relief valve is configured to supply the second fluid to the DCV in response to a pressure within the sensing line exceeding a shift pressure to move the DCV from the first position to the second position.

Another embodiment relates to a pump system. The pump system includes a housing, a first piston, a second piston, a

2

directional control valve (DCV), and a relief valve. The housing defines a first internal volume and a second internal volume. The first piston is positioned to separate the first internal volume into a first chamber and a second chamber. The second piston is positioned to separate the second internal volume into a third chamber and a fourth chamber. The first chamber and the third chamber are configured to selectively receive a first fluid from a first fluid source. The second chamber and the fourth chamber are configured to selectively receive a second fluid from a second fluid source. The DCV is fluidly coupled to the second chamber and the fourth chamber. The DCV is repositionable between (a) a first position where the DCV is configured to fluidly couple the second chamber to the second fluid source and (b) a second position where the DCV is configured to fluidly couple the fourth chamber to the second fluid source. The relief valve is fluidly coupled to the second chamber through a sensing line. The relief valve is configured to supply the second fluid to the DCV in response to a pressure within the sensing line exceeding a shift pressure to move the DCV from the first position to the second position.

Another embodiment relates to a pump system. The pump system includes a housing defining a first internal volume and a second internal volume, a first piston positioned to separate the first internal volume into a first chamber and a second chamber, a second piston positioned to separate the second internal volume into a third chamber and a fourth chamber, a directional control valve (DCV) fluidly coupled to the second chamber and the fourth chamber, a first relief valve fluidly coupled to the DCV via a first control line and the second chamber via a first sensing line, a first orifice positioned along the first sensing line, a second relief valve fluidly coupled to the DCV via a second control line and the fourth chamber via a second sensing line, and a second orifice positioned along the second sensing line.

This summary is illustrative only and is not intended to be in any way limiting. Other aspects, inventive features, and advantages of the devices or processes described herein will become apparent in the detailed description set forth herein, taken in conjunction with the accompanying figures, wherein like reference numerals refer to like elements.

BRIEF DESCRIPTION OF THE FIGURES

FIG. 1 is a front perspective view of a pump, according to an exemplary embodiment;

FIG. 2 is a front section view of the pump of FIG. 1 with a piston assembly in a center position;

FIG. 3 is a left section view of the pump of FIG. 1;

FIG. 4 is a top section view of the pump of FIG. 1 with the piston assembly in an extreme left position;

FIG. 5 is a left side view of a first piston of the piston assembly of FIG. 4;

FIG. 6 is a right side view of a second piston of the piston assembly of FIG. 4;

FIG. 7 is a right section view of the first piston of FIG. 5;

FIG. 8 is a left section view of the second piston of FIG. 6;

FIG. 9 is a front section view of the pump of FIG. 1 with the piston assembly in the extreme left position;

FIG. 10 is a front section view of the pump of FIG. 1 with the piston assembly in an extreme right position;

FIGS. 11-16 are schematic views of a pump system including the pump of FIG. 1 at various stages throughout a pumping cycle;

FIG. 17 is a section view of a pump controller of a pump system, according to an exemplary embodiment;

3

FIG. 18 is a schematic view of a pump system including the pump of FIG. 1, according to another exemplary embodiment;

FIGS. 19 and 20 are partial section views of a pump, according to various exemplary embodiments;

FIG. 21 is a front section view of a pump, according to another exemplary embodiment;

FIG. 22 is a front section view of a pump, according to another exemplary embodiment;

FIG. 23 is a front perspective view of an axle assembly for a vehicle, according to an exemplary embodiment;

FIG. 24 is a side perspective view of the axle assembly of FIG. 23; and

FIG. 25 is a schematic view of a suspension control assembly for the vehicle of FIG. 23, according to an exemplary embodiment.

DETAILED DESCRIPTION

Before turning to the figures, which illustrate certain exemplary embodiments in detail, it should be understood that the present disclosure is not limited to the details or methodology set forth in the description or illustrated in the figures. It should also be understood that the terminology used herein is for the purpose of description only and should not be regarded as limiting.

Referring generally to the figures, a pump system includes a pump and a pump controller. Specifically, the pump is a reciprocating piston pump including two pistons that are coupled by a rod to form a piston assembly. The first piston divides a first volume of a housing into a first outer chamber and a first inner chamber. The second piston divides a second volume of the housing into a second outer chamber and a second inner chamber. Each outer chamber is fluidly coupled to an inlet check valve and an outlet check valve. Accordingly, as the piston assembly reciprocates, a first fluid (e.g., nitrogen) is drawn into one outer chamber through the corresponding inlet check valve and expelled from the other outer chamber through the corresponding outlet check valve, with the direction of flow for each chamber being based on a movement direction of the piston assembly.

The pump controller provides a second fluid (e.g., hydraulic oil) alternately to each inner chamber to cause the piston assembly to reciprocate. In one embodiment, the pump controller includes a two-position directional control valve that controls which inner chamber is being supplied with the second fluid. A pair of relief valves change the position of the directional control valve in response to a pressure within one of the inner chambers exceeding a shift pressure. A pair of orifices are positioned to delay the flow of fluid through the relief valve, which in turn delays the shifting of the position of the directional control valve. This permits the pressure supplied to the inner chambers to increase above the shift pressure, beneficially increasing the maximum output pressure of the pump relative to an identical pump without the orifices. Additionally, in some arrangements, the first fluid is pressurized prior to entering the outer chambers. Due to the positions of the outer chambers relative to the piston assembly, this further increases the maximum output pressure of the pump.

Pump Assembly

Referring to FIG. 1, a fluid conveyor (e.g., a high-pressure gas (HPG) piston pump, a pump assembly, a compressor assembly, etc.) is shown as pump 100 according to an exemplary embodiment. The pump 100 is configured to

4

intake a fluid (e.g., gas, liquid, etc.) and expel the fluid at a higher pressure. In some embodiments, the pump 100 is configured to increase the pressure of an inert gas (e.g., nitrogen, argon, etc.).

As shown in FIGS. 1 and 2, the pump 100 includes a central portion or divider, shown as main body 102, coupled to a pair of piston assemblies, shown as end assemblies 104 and 106. The end assemblies 104 and 106 may be substantially similar to one another, except as otherwise specified herein. The main body 102 is substantially centered between the end assembly 104 and the end assembly 106. The main body 102, the end assembly 104, and the end assembly 106 are substantially centered about a longitudinal axis, shown as actuation axis 108.

The end assembly 104 includes an end portion, shown as cap 110, oriented substantially perpendicular to the actuation axis 108. Extending between the main body 102 and the cap 110 is a tubular member (e.g., a piston bore sleeve), shown as piston tube 112. A pair of annular seals, shown as o-rings 114, extend between the cap 110 and the piston tube 112 and between the piston tube 112 and the main body 102, respectively. The o-rings 114 substantially seal the connections between the cap 110, the piston tube 112, and the main body 102, such that a first sealed volume is formed between the cap 110, the piston tube 112, and the main body 102. A series of threaded rods 115 extend longitudinally between and engage the cap 110 and the main body 102, coupling the cap 110 to the main body 102. In one embodiment, the threaded rods 115 are arranged outside of the piston tube 112. The piston tube 112 is captured between the cap 110 and the main body 102. In some embodiments, the piston tube 112 is located relative to the cap 110 and the main body 102 by engagement with a shoulder or a protrusion of the cap 110 and/or the main body 102.

A first piston or cylindrical member, shown as piston 116, is positioned within the first sealed volume of the end assembly 104. The piston 116 sealingly engages an inner surface of the piston tube 112, dividing the first sealed volume into a first chamber or volume, shown as outer chamber 120, and a second chamber or volume, shown as inner chamber 122. The piston 116 is substantially centered about the actuation axis 108. The piston 116 is slidably coupled to the piston tube 112 such that the piston 116 can translate along the actuation axis 108 to vary the volumes of the outer chamber 120 and the inner chamber 122.

Referring to FIGS. 2-4, the cap 110 defines a first passage, shown as inlet passage 130, and a second passage, shown as outlet passage 132, extending therethrough. The inlet passage 130 and the outlet passage 132 are fluidly coupled to the outer chamber 120. A first directionally-dependent flow control device or check valve assembly, shown as inlet check valve 134, is coupled to the cap 110 and positioned along the inlet passage 130. The inlet check valve 134 is configured to permit flow (e.g., minimally obstructed flow) of fluid through the inlet passage 130 and into the outer chamber 120. The inlet check valve 134 is configured to limit (e.g., prevent) flow of fluid through the inlet passage 130 and out of the outer chamber 120. A second directionally-dependent flow control device or check valve assembly, shown as outlet check valve 136, is coupled to the cap 110 and positioned along the outlet passage 132. The outlet check valve 136 is configured to permit flow (e.g., minimally obstructed flow) of fluid through the outlet passage 132 and out of the outer chamber 120. The outlet check valve 136 is configured to limit (e.g., prevent) flow of fluid through the outlet passage 132 and into the outer chamber 120.

5

A third passage, shown as passage A, is defined by the main body 102 and extends therethrough. The passage A is fluidly coupled to the inner chamber 122. A portion of the inlet passage 130 and a portion of the outlet passage 132 adjacent the outer chamber 120 extend substantially parallel to the actuation axis 108. A portion of the passage A adjacent the inner chamber 122 extends substantially parallel to the actuation axis 108.

The end assembly 106 includes an end portion, shown as cap 210, oriented substantially perpendicular to the actuation axis 108. Extending between the main body 102 and the cap 210 is a tubular member (e.g., a piston bore sleeve), shown as piston tube 212. A pair of annular seals, shown as o-rings 214, extend between the cap 210 and the piston tube 212 and between the piston tube 212 and the main body 102, respectively. The o-rings 214 substantially seal the connections between the cap 210, the piston tube 212, and the main body 102, such that a second sealed volume is formed between the cap 210, the piston tube 212, and the main body 102. A series of threaded rods 215 extend longitudinally between and engage the cap 210 and the main body 102, coupling the cap 210 to the main body 102. In one embodiment, the threaded rods 215 are arranged outside of the piston tube 212. The piston tube 212 is captured between the cap 210 and the main body 102. In some embodiments, the piston tube 212 is located relative to the cap 210 and the main body 102 by engagement with a shoulder or a protrusion of the cap 210 and/or the main body 102.

A second piston or cylindrical member, shown as piston 216, is positioned within the second sealed volume of the end assembly 106. The piston 216 sealingly engages an inner surface of the piston tube 212, dividing the second sealed volume into a third chamber or volume, shown as outer chamber 220, and a fourth chamber or volume, shown as inner chamber 222. The inner chamber 122 is separated from the inner chamber 222 by the main body 102. The piston 216 is substantially centered about the actuation axis 108. The piston 216 is slidably coupled to the piston tube 212 such that the piston 216 can translate along the actuation axis 108 to vary the volumes of the outer chamber 220 and the inner chamber 222.

The cap 210 defines a first passage, shown as inlet passage 230, and a second passage, shown as outlet passage 232, extending therethrough. The inlet passage 230 and the outlet passage 232 are fluidly coupled to the outer chamber 220. A first directionally-dependent flow control device or check valve assembly, shown as inlet check valve 234, is coupled to the cap 210 and positioned along the inlet passage 230. The inlet check valve 234 is configured to permit flow (e.g., minimally obstructed flow) of fluid through the inlet passage 230 and into the outer chamber 220. The inlet check valve 234 is configured to limit (e.g., prevent) flow of fluid through the inlet passage 230 and out of the outer chamber 220. A second directionally-dependent flow control device or check valve assembly, shown as outlet check valve 236, is coupled to the cap 210 and positioned along the outlet passage 232. The outlet check valve 236 is configured to permit flow (e.g., minimally obstructed flow) of fluid through the outlet passage 232 and out of the outer chamber 220. The outlet check valve 236 is configured to limit (e.g., prevent) flow of fluid through the outlet passage 232 and into the outer chamber 220.

A third passage, shown as passage B, is defined by the main body 102 and extends therethrough. The passage B is fluidly coupled to the inner chamber 222. A portion of the inlet passage 230 and a portion of the outlet passage 232 adjacent the outer chamber 220 extend substantially parallel

6

to the actuation axis 108. A portion of the passage B adjacent the inner chamber 222 extends substantially parallel to the actuation axis 108.

The cap 110, the piston tube 112, the main body 102, the piston tube 212, and the cap 210 are part of a housing 240. The housing 240 defines the first sealed volume and the second sealed volume that are divided into the outer chamber 120, the inner chamber 122, the outer chamber 220, and the inner chamber 222 by the piston 116, the piston 216, and the main body 102.

An aperture, shown as piston rod aperture 250, extends longitudinally through the main body 102 between the inner chamber 122 and the inner chamber 222. Specifically, the piston rod aperture 250 is substantially centered about the actuation axis 108. A shaft, rod, or coupler, shown as piston rod 252, extends through the piston rod aperture 250. A first end of the piston rod 252 is coupled (e.g., fixedly, through a threaded engagement, etc.) to the piston 116. A second, opposite end of the piston rod 252 is coupled (e.g., fixedly, through a threaded engagement, etc.) to the piston 216. A pair of annular seals, shown as o-rings 254, are coupled to the main body 102 along an inner surface of the piston rod aperture 250. The o-rings 254 engage the piston rod 252 and the main body 102, each creating a seal that fluidly decouples the inner chamber 122 from the inner chamber 222.

The piston rod 252 fixedly couples the piston 116 to the piston 216 to form a piston assembly 260 (i.e., the piston assembly 260 includes the piston 116, the piston 216, and the piston rod 252). Accordingly, the movement of each piston 116, 216 along the actuation axis 108 is equal (i.e., in the same direction and the same distance). By way of example, when the piston 116 moves 0.25 inches to the left, the piston 216 also moves 0.25 inches to the left. By way of another example, when the piston 216 moves 1 inch to the right, the piston 116 also moves 1 inch to the right.

Referring to FIGS. 5 and 6, front views of the piston 116 and the piston 216 are shown. In this embodiment, the piston 116 is substantially cylindrical and has a constant outer radius R_{P1} along the actuation axis 108. The area A_{P1O} of the outer surface of the piston 116 (i.e., the surface of the piston 116 that is exposed to the outer chamber 120) can be calculated using the following equation:

$$A_{P1O} = \pi R_{P1}^2 \quad (1)$$

Similarly, the piston 216 is substantially cylindrical and has a constant outer radius R_{P2} along the actuation axis 108. The area A_{P2O} of the outer surface of the piston 216 (i.e., the surface of the piston 216 that is exposed to the outer chamber 220) can be calculated using the following equation:

$$A_{P2O} = \pi R_{P2}^2 \quad (2)$$

Referring to FIGS. 7 and 8, rear views of the piston 116 and the piston 216 are shown. In this embodiment, the piston rod 252 is substantially cylindrical and has a constant outer radius R_R along the actuation axis 108. The piston rod 252 has a cross-sectional area A_R where the piston rod 252 meets the piston 116 and the piston 216. An area A_{P1I} of the inner surface of the piston 116 (i.e., the surface of the piston 116 that is exposed to the inner chamber 122) and an area A_{P2I} of the inner surface of the piston 216 (i.e., the surface of the piston 216 that is exposed to the inner chamber 222) can be calculated as follows:

$$A_{P1I} = A_{P1O} - A_R = \pi R_{P1}^2 - \pi R_R^2 \quad (3)$$

$$A_{P2I} = A_{P2O} - A_R = \pi R_{P2}^2 - \pi R_R^2 \quad (4)$$

The areas of the inner surfaces are smaller than the areas of the corresponding outer surfaces due to the presence of the piston rod **252** on the inner surfaces.

Referring to FIG. 2, as the piston assembly **260** travels along the actuation axis **108**, the volumes of each inner volume and outer volume vary. By way of example, when the piston **116** travels in a first direction (e.g., to the left as shown in FIG. 2), the outer chamber **120** decreases in volume and the inner chamber **122** increases in volume. Due to the coupling of the piston **116** and the piston **216** by the piston rod **252**, the piston **216** also travels in the first direction such that the outer chamber **220** increases in volume and the inner chamber **222** decreases in volume. By way of another example, when the piston **116** travels in a second direction (e.g., to the right as shown in FIG. 2), the outer chamber **120** increases in volume and the inner chamber **122** decreases in volume. Due to the coupling of the piston **116** and the piston **216**, the piston **216** also travels in the second direction such that the outer chamber **220** decreases in volume and the inner chamber **222** increases in volume.

The position of the piston assembly **260** along the actuation axis **108** is referred to herein as position X, where X=0 when the piston assembly **260** is centered within the pump **100** and where X increases when the piston assembly **260** moves to the left as shown in FIG. 2. The outer chamber **120**, the inner chamber **122**, the outer chamber **220**, and the inner chamber **222** have a volume V_1 , a volume V_2 , a volume V_3 , and a volume V_4 , respectively. These volumes depend upon the areas of each corresponding piston surface and the position X of the piston assembly **260**. Accordingly, the rate of change of each volume (e.g., $\dot{V}_1$, etc.) may be calculated as follows:

$$\dot{V}_1 = -A_{P1O} \times \dot{X} \quad (5)$$

$$\dot{V}_2 = A_{P1I} \times \dot{X} \quad (6)$$

$$\dot{V}_3 = A_{P2O} \times \dot{X} \quad (7)$$

$$\dot{V}_4 = A_{P2I} \times \dot{X} \quad (8)$$

where X is the rate of change of the position X. Because each inner volume and corresponding outer volume are positioned on opposite sides of the corresponding piston, the corresponding rates of change (e.g., the rate of change of the volume V_1 and the volume V_2) have opposite signs (i.e., when one increases, the other decreases).

The fluid within each chamber has a corresponding pressure. Specifically, the fluids within the outer chamber **120**, the inner chamber **122**, the outer chamber **220**, and the inner chamber **222** have corresponding pressures P_1 , P_2 , P_3 , and P_4 , respectively. The forces on the piston assembly **260** are dependent on the pressures within the chambers and the corresponding exposed areas of the pistons. Specifically, the sum of the forces on the piston assembly **260** along the actuation axis **108** (i.e., ΣF_x) can be calculated as follows:

$$\Sigma F_x = -A_{P1O}P_1 + A_{P1I}P_2 + A_{P2O}P_3 - A_{P2I}P_4 = Ma_x \quad (9)$$

where M is the mass of the piston assembly **260** and a_x is the acceleration of the piston assembly **260** in the positive X direction.

The piston assembly **260** may driven to reciprocate (e.g., by selectively applying pressurized fluid to different chambers). In some embodiments, due to this reciprocation, the pump **100** is configured to draw in fluid at a low pressure (e.g., atmospheric pressure, above atmospheric pressure, below atmospheric pressure) through the inlet passage **130** and the inlet passage **230** (e.g., from the same fluid source,

from different fluid sources, etc.) and expel the fluid at a higher pressure through the outlet passage **132** and the outlet passage **232** (e.g., to the same chamber, to different chambers, etc.).

Referring to FIGS. 2, 9, and 10, a cycle of the pump **100** is shown and described according to an exemplary embodiment. Although the piston assembly **260** is shown starting at a particular position, it should be understood that the piston assembly **260** may begin at any position and may begin by travelling in either direction.

As shown in FIG. 2, the piston assembly **260** begins in a central position (e.g., X=0). A pressurized fluid is supplied to the inner chamber **122** through the passage A, forcing the piston assembly **260** to move in the positive X direction (e.g., to the left as shown in FIG. 2). This decreases the volume V_1 of the outer chamber **120** and the volume V_4 of the inner chamber **222** and increases the volume V_2 of the inner chamber **122** and the volume V_3 of the outer chamber **220**.

As the volume V_1 decreases, the pressure P_1 increases, and the outlet check valve **136** opens to permit pressurized fluid to exit through the outlet passage **132**. Due to the pressure increase, the inlet check valve **134** closes, preventing fluid from flowing out through the inlet passage **130**. As the volume V_3 increases, the pressure P_3 decreases, and the inlet check valve **234** opens to permit fluid to enter into the outer chamber **220** through the inlet passage **230**. Due to the pressure decrease, the outlet check valve **236** closes, preventing fluid from flowing out through the outlet passage **232**. Fluid may additionally flow out of the inner chamber **222** through the passage B. The piston assembly **260** may move in the first direction until the force of the fluid within the volume V_2 decreases (e.g., due to control by a control system), until the force of the fluid within another volume overcomes the force of the fluid within the volume V_2 , or until the piston assembly **260** engages a hard stop (e.g., the cap **110** as shown in FIG. 9).

Subsequently, a pressurized fluid may be supplied to the inner chamber **222** through the passage B, forcing the piston assembly **260** to move in the negative X direction (e.g., to the right as shown in FIG. 2). This increases the volume V_1 of the outer chamber **120** and the volume V_4 of the inner chamber **222** and decreases the volume V_2 of the inner chamber **122** and the volume V_3 of the outer chamber **220**.

As the volume V_1 increases, the pressure P_1 decreases, and the inlet check valve **134** opens to permit fluid to enter into the outer chamber **120** through the inlet passage **130**. Due to the pressure decrease, the outlet check valve **136** closes, preventing fluid from flowing out through the outlet passage **132**. As the volume V_3 decreases, the pressure P_3 increases, and the outlet check valve **236** opens to permit pressurized fluid to exit the outlet passage **232**. Due to the pressure increase, the inlet check valve **234** closes, preventing fluid from flowing out through the inlet passage **230**. Fluid may additionally flow out of the inner chamber **122** through the passage A. The piston assembly **260** may move in the second direction until the force of the fluid within the volume V_4 decreases (e.g., due to control by a control system), until the force of the fluid within another chamber overcomes the force of the fluid within the volume V_4 , or until the piston assembly **260** engages a hard stop (e.g., the cap **210** as shown in FIG. 10). The movement direction of the piston assembly **260** may once again be reversed, and the cycle may be repeated.

Referring to FIGS. 2-4, 9, and 10, the due to the positions of the inlet passage **130** and the outlet passage **132** within the cap **110**, the inlet passage **130**, the outlet passage **132**, the

inlet check valve **134**, and the outlet check valve **136** are all fluidly coupled to the outer chamber **120** throughout the entire range of motion of the piston assembly **260**. Specifically, the inlet passage **130** and the outlet passage **132** face an end of the piston **116** such that the inlet passage **130** and the outlet passage **132** are not covered by a circumferential surface of the piston **116**. Accordingly, the pump **100** can utilize the full stroke of the piston assembly **260** to pump fluid. Similarly, the inlet passage **230**, the outlet passage **232**, the inlet check valve **234**, and the outlet check valve **236** are fluidly coupled to the outer chamber **220** throughout the entire range of motion of the piston assembly **260**.

Pump Control

Referring to FIG. **11**, the pump **100** is fluidly coupled to and controlled by a mechanical control system, valve assembly, or automatic pump controller, shown as pump controller **300**, to form a pump system **302**. The pump controller **300** is configured to repeatedly switch between (a) supplying pressurized fluid (e.g., pressurized hydraulic fluid) to a first chamber of the pump **100** and draining fluid from a second chamber of the pump **100** and (b) supplying pressurized fluid to the second chamber and draining fluid from the first chamber. By quickly repeating this process, the pump controller **300** causes the piston assembly **260** to oscillate or reciprocate such that the pump **100** pumps pressurized fluid out of the remaining third and fourth chambers of the pump **100**.

In the embodiment shown in FIG. **11**, the pump controller **300** is configured to supply a pressurized fluid to the inner chamber **122** and the inner chamber **222** to control movement of the piston assembly **260**. In some embodiments, the fluid supplied by the pump controller **300** is hydraulic oil. As shown, the pump controller **300** is a mechanical controller that automatically changes the direction of movement of the piston assembly **260** based on the pressures P_2 and P_4 within the inner chamber **122** and the inner chamber **222**, respectively. Specifically, when the pressure within one of the chambers exceeds a threshold pressure, the pump controller **300** changes the direction of movement of the piston assembly **260**.

The pump controller **300** defines a first inlet, outlet, connection, passage, conduit, or port, shown as supply passage **310**, that is fluidly coupled to a high pressure fluid source, shown as supply **P**, and a second inlet, outlet, connection, passage, conduit, or port, shown as tank passage **312**, that is fluidly coupled to a low pressure fluid sink, shown as tank **T**. By way of example, the supply **P** may be a hydraulic pump that is supplied with hydraulic fluid from the tank **T**.

The supply passage **310** and the tank passage **312** are fluidly coupled to a flow control device (e.g., a spool valve, etc.) or directional control valve (DCV), shown as DCV **320**. The DCV **320** is fluidly coupled to the passage **A** and the passage **B** (e.g., directly, indirectly through another conduit, etc.). As shown, the DCV **320** is a four-port, two-position valve. Specifically, the ports are fluidly coupled to the supply passage **310**, the tank passage **312**, the passage **A**, and the passage **B**, respectively. The DCV **320** is configured to control the flow between the ports based on a position of the valve (e.g., a position of a spool within the valve). In a first position, shown in FIG. **11**, the DCV **320** fluidly couples the supply passage **310** to the passage **A** and fluidly couples the tank passage **312** to the passage **B**. In a second position, shown in FIG. **13**, the DCV **320** fluidly couples the supply passage **310** to the passage **B** and fluidly couples the tank

passage **312** to the passage **A**. In some embodiments, the DCV **320** has additional positions (e.g., a center position in which no flow is permitted through the DCV **320**). In some embodiments, the DCV **320** is a single valve. In other embodiments, the DCV **320** is a combination of multiple valves. As shown, the position of the DCV **320** is hydraulically controlled. In other embodiments, the DCV **320** is otherwise controlled (e.g., electronically, pneumatically, etc.).

A first flow control device or control device, shown as relief valve **330**, is fluidly coupled to the passage **A** through a first conduit, shown as sensing line **332**. In some embodiments, a flow control device or flow resistor (e.g., an orifice, a valve, etc.), shown as orifice **334**, is fluidly coupled between the sensing line **332** and the passage **A**. The orifice **334** is configured to restrict flow of fluid into the sensing line **332** from the passage **A**. The relief valve **330** is fluidly coupled to the DCV **320** through a second conduit, shown as control line **336**.

Similarly, a second flow control device or control device, shown as relief valve **340**, is fluidly coupled to the passage **B** through a second conduit, shown as sensing line **342**. In some embodiments, a flow control device or flow resistor (e.g., an orifice, a valve, etc.), shown as orifice **344**, is fluidly coupled between the sensing line **342** and the passage **B**. The orifice **344** is configured to restrict flow of fluid into the sensing line **342** from the passage **B**. In some embodiments, the sensing line **332** and/or the sensing line **342** are relatively low flow lines (e.g., as compared to the passage **A** or the passage **B**).

The relief valve **330** is configured to selectively couple the sensing line **332** to a first side of the DCV **320**. When pressurized fluid is supplied to the passage **A**, it passes through the orifice **334** and into the sensing line **332**. The relief valve **330** is biased into a closed position. As shown, the relief valve **330** is biased into a closed position by a spring that has a variable preload force. In other embodiments, the relief valve **330** is otherwise biased (e.g., by a spring with a fixed preload force, pneumatically, hydraulically, etc.). The relief valve **330** is controlled to move toward an open position by the pressurized fluid within the sensing line **332**. Accordingly, when the pressure within the sensing line **332** exceeds a threshold pressure (i.e., a shift pressure for the relief valve **330**), the force of the pressurized fluid overcomes the preload force of the spring, opening the relief valve **330**. The shift pressure may be controlled by varying the preload force on the spring. When the relief valve **330** is opened, the pressurized fluid within the sensing line **332** passes through the relief valve **330** and the control line **336** to the first side of the DCV **320**. The pressurized fluid at the first side forces the DCV **320** to move toward the second position (e.g., as shown in FIG. **13**).

The relief valve **340** is configured to selectively couple the sensing line **342** to a second side of the DCV **320** opposite the first side. When pressurized fluid is supplied to the passage **B**, it passes through the orifice **344** and into the sensing line **342**. The relief valve **340** is biased into a closed position. As shown, the relief valve **340** is biased into a closed position by a spring that has a variable preload force. In other embodiments, the relief valve **340** is otherwise biased (e.g., by a spring with a fixed preload force, pneumatically, hydraulically, etc.). The relief valve **340** is controlled to move toward an open position by the pressurized fluid within the sensing line **342**. Accordingly, when the pressure within the sensing line **342** exceeds a threshold pressure (i.e., a shift pressure for the relief valve **340**), the force of the pressurized fluid overcomes the preload force of

11

the spring, opening the relief valve **340**. The shift pressure may be controlled by varying the preload force on the spring. When the relief valve **340** is opened, the pressurized fluid within the sensing line **342** passes through the relief valve **340** and the control line **346** to the second side of the DCV **320**. This pressurized fluid at the second side forces the DCV **320** to move toward the first position (e.g., as shown in FIG. **11**).

The orifice **334** and the orifice **344** are configured to resist the flow of fluid therethrough. Accordingly, when the pressure within the passage A or the passage B is changing, a pressure differential is created across the orifice **334** or the orifice **344**, respectively. The pressure within the sensing line **332** immediately downstream of the orifice **334** is referred to herein as pressure P_5 , and the pressure within the sensing line **342** immediately downstream of the orifice **344** is referred to herein as pressure P_6 . In other embodiments, the orifice **334** is positioned in a different location between the passage A and the DCV **320** (e.g., within the relief valve **330**, along the control line **336**, etc.). Similarly, the orifice **344** may be positioned in a different location between the passage B and the DCV **320** (e.g., within the relief valve **340**, along the control line **346**, etc.).

Referring to FIGS. **11-16**, each stage of a full control cycle of the pump system **302** is shown and described. It should be understood that, although one stage is shown as the first step in this process, the pump system **302** may begin this cycle at any point throughout the process. In other embodiments, one or more stages are removed, reordered, or added to the cycle. Additionally, the fluid within the pump controller **300** and the inner chambers is described as being hydraulic oil, and the fluid within the outer chambers is described as being nitrogen gas. In other embodiments, one or both fluids are a different type of fluid.

The pressures within each chamber are described herein as reaching certain pressure thresholds or pressure ranges. As used herein, “system pressure” or “supply pressure” refers to the maximum pressure provided by the supply P (e.g., when no flow is occurring). In some embodiments, the supply pressure is 3250 psi. As used herein, “shift pressure” refers to the pressure at which the relief valve **330** and the relief valve **340** are configured to open. In some embodiments, the shift pressure is 3100 psi. As used herein, “filling pressure” is the pressure within the passage A or the passage B as the piston assembly **260** is moving. The filling pressure may be a single pressure or a range, and may depend upon the resistance of the piston assembly **260** to movement.

Pumping Cycle—Stage 1

Referring to FIG. **11**, in stage 1 of the cycle, the DCV **320** is in the first position and the piston assembly **260** is somewhere between the extreme left position (i.e., the farthest position reached by the piston assembly **260** in the positive X direction) and the extreme right position (i.e., the farthest position reached by the piston assembly **260** in the negative X direction). The pressures within each chamber are described in Table 1 below:

TABLE 1

Pressures of Each Chamber in Stage 1		
Pressure	Location	Pressure Value
P_1	Outer Chamber 120	High Pressure/Nitrogen Compression

12

TABLE 1-continued

Pressures of Each Chamber in Stage 1		
Pressure	Location	Pressure Value
P_2	Inner Chamber 122/Passage A	Filling Pressure
P_3	Outer Chamber 220	Low Pressure/Nitrogen Intake
P_4	Inner Chamber 222/Passage B	Tank Pressure
P_5	Sensing Line 332	Variable
P_6	Sensing Line 342	Variable

In stage 1, high pressure hydraulic oil is supplied from the supply P through the DCV **320** and the passage A and into the inner chamber **122**. Similarly, hydraulic oil from the inner chamber **222** is permitted to drain through the passage B and the DCV **320** to the tank T. Pressurized hydraulic oil enters and expands the inner chamber **122**, forcing the piston assembly **260** in the positive X direction. The volume V_1 of the outer chamber **120** is reduced, compressing the nitrogen therein and forcing nitrogen out through the outlet passage **132**. The volume V_3 of the outer chamber **220** increases, drawing low pressure nitrogen into the outer chamber **220**. The filling pressure within the passage A may be less than the system pressure provided by the supply P, as the inner chamber **122** is free to expand instead of building pressure. The pressures in the sensing lines may vary throughout stage 1. By way of example, the pressure P_5 may begin relatively low and increase gradually (e.g., due to exposure to the filling pressure through the orifice **334**), and the pressure P_6 may begin relatively high and decrease gradually (e.g., due to exposure to the tank pressure through the orifice **344**).

Pumping Cycle—Stage 2

Referring to FIG. **12**, in stage 2 of the cycle, the DCV **320** remains in the first position and the piston assembly **260** is at or near the extreme left position (i.e., the farthest position reached by the piston assembly **260** in the positive X direction). As shown herein, the piston assembly **260** is bottomed out (e.g., contacting the cap **110**) in the extreme left position. In other embodiments, the extreme left position is elsewhere. The pressures within each chamber are described in Table 2 below:

TABLE 2

Pressures of Each Chamber in Stage 2		
Pressure	Location	Pressure Value
P_1	Outer Chamber 120	High Pressure/Nitrogen Compression
P_2	Inner Chamber 122/Passage A	Shift Pressure
P_3	Outer Chamber 220	Low Pressure/Nitrogen Intake
P_4	Inner Chamber 222/Passage B	Tank Pressure
P_5	Sensing Line 332	Below Shift Pressure
P_6	Sensing Line 342	Below Shift Pressure

In stage 2, the piston assembly **260** experiences resistance to further movement in the positive X direction (e.g., due to an increased nitrogen pressure within the outer chamber **120**, due to the piston assembly **260** encountering a hard stop, etc.). Accordingly, the pressure P_2 increases until it reaches the shift pressure. Due to the delay in pressure communication introduced by the orifice **334**, the pressure P_5 has not yet reached the shift pressure, and thus the relief valve **330** has not moved to the open position.

Pumping Cycle—Stage 3

Referring again to FIG. **12**, in stage 3 of the cycle, the DCV **320** switches to the second position and the piston

13

assembly **260** remains at or near the extreme left position (i.e., the farthest position reached by the piston assembly **260** in the positive X direction). The pressures within each chamber are described in Table 3 below:

TABLE 3

Pressures of Each Chamber in Stage 3		
Pressure	Location	Pressure Value
P ₁	Outer Chamber 120	High Pressure/Nitrogen Compression
P ₂	Inner Chamber 122/Passage A	Supply Pressure
P ₃	Outer Chamber 220	Low Pressure/Nitrogen Intake
P ₄	Inner Chamber 222/Passage B	Tank Pressure
P ₅	Sensing Line 332	Shift Pressure
P ₆	Sensing Line 342	Below Shift Pressure

In stage 3, the piston assembly **260** is located in the extreme left position. Due to the delay introduced by the orifice **334**, the pressure P₂ increases until reaching the supply pressure prior to activation of the relief valve **330**. After the pressure P₂ has increased to the point of reaching the supply pressure, the pressure P₅ reaches the shift pressure, causing the relief valve **330** to move to the open position. This permits the pressurized hydraulic oil to pass through the relief valve **330** to the first end of the DCV **320**, forcing the DCV **320** into the second position and beginning movement of the piston assembly **260** in the negative X direction.

Pumping Cycle—Stage 4

Referring to FIG. 14, in stage 4 of the cycle, the DCV **320** is in the second position and the piston assembly **260** is somewhere between the extreme left position (i.e., the farthest position reached by the piston assembly **260** in the positive X direction) and the extreme right position (i.e., the farthest position reached by the piston assembly **260** in the negative X direction). The pressures within each chamber are described in Table 4 below:

TABLE 4

Pressures of Each Chamber in Stage 4		
Pressure	Location	Pressure Value
P ₁	Outer Chamber 120	Low Pressure/Nitrogen Intake
P ₂	Inner Chamber 122/Passage A	Tank Pressure
P ₃	Outer Chamber 220	High Pressure/Nitrogen Compression
P ₄	Inner Chamber 222/Passage B	Filling Pressure
P ₅	Sensing Line 332	Variable
P ₆	Sensing Line 342	Variable

In stage 4, high pressure hydraulic oil is supplied from the supply P through the DCV **320** and the passage B and into the inner chamber **222**. Similarly, hydraulic oil from the inner chamber **122** is permitted to drain through the passage A and the DCV **320** to the tank T. Pressurized hydraulic oil enters and expands the inner chamber **222**, forcing the piston assembly **260** in the negative X direction. The volume V₃ of the outer chamber **220** is reduced, compressing the nitrogen therein and forcing nitrogen out through the outlet passage **232**. The volume V₁ of the outer chamber **120** increases, drawing low pressure nitrogen into the outer chamber **120**. The filling pressure within the passage B may be less than the system pressure provided by the supply P, as the inner chamber **222** is free to expand instead of building pressure.

14

The pressures in the sensing lines may vary throughout stage 4. By way of example, the pressure P₅ may begin relatively high and decrease gradually (e.g., due to exposure to the tank pressure through the orifice **334**), and the pressure P₆ may begin relatively low and increase gradually (e.g., due to exposure to the filling pressure through the orifice **344**).

Pumping Cycle—Stage 5

Referring to FIG. 15, in stage 5 of the cycle, the DCV **320** remains in the second position and the piston assembly **260** is at or near the extreme right position (i.e., the farthest position reached by the piston assembly **260** in the negative X direction). As shown herein, the piston assembly **260** is bottomed out (e.g., contacting the cap **210**) in the extreme right position. In other embodiments, the extreme right position is elsewhere. The pressures within each chamber are described in Table 5 below:

TABLE 5

Pressures of Each Chamber in Stage 5		
Pressure	Location	Pressure Value
P ₁	Outer Chamber 120	Low Pressure/Nitrogen Intake
P ₂	Inner Chamber 122/Passage A	Tank Pressure
P ₃	Outer Chamber 220	High Pressure/Nitrogen Compression
P ₄	Inner Chamber 222/Passage B	Shift Pressure
P ₅	Sensing Line 332	Below Shift Pressure
P ₆	Sensing Line 342	Below Shift Pressure

In stage 5, the piston assembly **260** experiences resistance to further movement in the negative X direction (e.g., due to an increased nitrogen pressure within the outer chamber **220**, due to the piston assembly **260** encountering a hard stop, etc.). Accordingly, the pressure P₄ increases until it reaches the shift pressure. Due to the delay in pressure communication introduced by the orifice **344**, the pressure P₆ has not yet reached the shift pressure, and thus the relief valve **340** has not moved to the open position.

Pumping Cycle—Stage 6

Referring again to FIG. 16, in stage 6 of the cycle, the DCV **320** switches back to the first position and the piston assembly **260** remains at or near the extreme right position (i.e., the farthest position reached by the piston assembly **260** in the negative X direction). The pressures within each chamber are described in Table 6 below:

TABLE 6

Pressures of Each Chamber in Stage 6		
Pressure	Location	Pressure Value
P ₁	Outer Chamber 120	Low Pressure/Nitrogen Intake
P ₂	Inner Chamber 122/Passage A	Tank Pressure
P ₃	Outer Chamber 220	High Pressure/Nitrogen Compression
P ₄	Inner Chamber 222/Passage B	Supply Pressure
P ₅	Sensing Line 332	Below Shift Pressure
P ₆	Sensing Line 342	Shift Pressure

In stage 6, the piston assembly **260** is located in the extreme right position. Due to the delay introduced by the orifice **344**, the pressure P₄ increases until reaching the supply pressure prior to activation of the relief valve **340**. After the pressure P₄ has increased to the point of reaching

15

the supply pressure, the pressure P_6 reaches the shift pressure, causing the relief valve **340** to move to the open position. This permits the pressurized hydraulic oil to pass through the relief valve **340** to the second end of the DCV **320**, forcing the DCV **320** back into the first position and beginning movement of the piston assembly **260** in the positive X direction. The pump system **302** may then enter back into stage 1 of the cycle and repeat the process.

The speed of the pump **100** (e.g., the reciprocation rate in Hz, the flow rate of nitrogen out of the pump **100**, etc.) is determined by the flow rate of fluid supplied by the supply P. If the flow rate of fluid from the supply P is increased, the pump **100** completes the pumping cycle faster. Accordingly, the flow rate of the pump **100** can be controlled by adding a flow control device between the supply P and the pump controller **300**, or by controlling the supply P itself.

Because fluid can be supplied to both the inner chamber **122** and the inner chamber **222**, the pump **100** is actively driven to throughout its entire range of motion. In other pumps, a piston is actively driven in a first direction, but relies on another force (e.g., from the compressed fluid, from a spring, etc.) to passively return it to its original position. When compared to these passively-returned pumps, the pump **100** can complete a pumping cycle more quickly and more reliably.

Beneficially, the orifice **334** and the orifice **344** increase the maximum output pressure of the pump **100** when compared to an identical embodiment where the orifice **334** and the orifice **344** are omitted. By delaying the shift of the DCV **320**, the pressure is permitted to increase beyond the shift pressure (e.g., 3100 psi), all of the way to the system pressure (e.g., 3250 psi). If the orifice **334** and the orifice **344** were to be omitted, the passage A would be directly fluidly coupled to the relief valve **330** and the passage B would be directly fluidly coupled to the relief valve **340**. Accordingly, the pressure P_2 would be substantially equal to the pressure P_5 and the pressure P_4 would be substantially equal to P_6 at all times. Accordingly, the DCV **320** would change positions immediately upon the pressure P_2 or the pressure P_4 reaching the shift pressure, such that stages 2 and 4 would effectively be omitted from the cycle.

Simply increasing the shift pressure to be closer to the system pressure would not provide the same benefit as the orifices **334**, **344**. Increasing the shift pressure would increase the pressure within the inner chamber **122** or the inner chamber **222** at the time of shifting. However, if the system pressure is not above the shift pressure, then the relief valve may not successfully open. If the system pressure is too close to the shift pressure, a fluctuation in system pressure (e.g., due to another load on the supply P) could prevent the relief valves **330**, **340** from opening. Accordingly, increasing the difference between the shift pressure and the system pressure ensures that this does not occur. The use of the orifices **334**, **344** permits the use of the full system pressure without the potential for an unsuccessful shift.

Referring to FIG. 17, a layout of the pump controller **300** is shown according to an exemplary embodiment. In this embodiment, the relief valves **330**, **340**, and the DCV **320** are all built into a housing **360**. The housing **360** at least partially defines the sensing line **332**, the control line **336**, the sensing line **342**, the control line **346**, the passage A, and the passage B. As shown, the orifice **334** is positioned between the sensing line **332** and the passage A, and the orifice **334** is positioned between the sensing line **342** and the passage B.

In other embodiments, the orifice **334** and/or the orifice **344** are otherwise located. By way of example, the orifice

16

334 may be positioned along the control line **336**. By way of another example, the orifice **334** may be positioned within the relief valve **330**. Moving the orifices within or downstream of the relief valves may cause the relief valves to shift more quickly (e.g., when the passage A or the passage B reaches the shift pressure. However, the orifices may still delay the pressurized fluid reaching the DCV **320**, and thus delay the shifting of the DCV **320** between the first and second positions.

Electronic Pump Control System

Referring to FIG. 18, a pump system **402** is shown as an alternative embodiment to the pump system **302**. The pump system **402** may be substantially similar to the pump system **302** except as otherwise specified herein. The pump system **402** includes an electronic pump controller, shown as pump controller **400**. The pump controller **400** is configured to alternately supply pressurized fluid to the passage A and the passage B, similar to the pump controller **300**. However, the pump controller **400** senses and controls the pressure electronically instead of mechanically.

The pump controller **400** defines a supply passage **410** and a tank passage **412** that fluidly couple the supply P and the tank T, respectively, to a directional control valve, shown as DCV **420**. The DCV **420** is substantially similar to the DCV **320** except the DCV **420** includes solenoids that move the DCV **420** between the first position and the second position in response to receiving electrical energy.

These solenoids are coupled to a processing circuit or control device, shown as controller **430**. The controller **430** is configured to control the solenoids to change the position of the DCV **420**. The controller **430** includes a processor **432** and a memory device, shown as memory **434**. The memory **434** may store control logic for execution by the processor **432**. The controller **430** is operatively coupled to a pair of pressure sensors (e.g., pressure transducers, etc.), shown as pressure sensor **440** and pressure sensor **442**. The pressure sensor **440** is fluidly coupled to the passage A. The pressure sensor **440** is configured to sense the pressure of the fluid within the passage A and provide a signal indicative of the pressure to the controller **430**. The pressure sensor **442** is fluidly coupled to the passage B. The pressure sensor **442** is configured to sense the pressure of the fluid within the passage B and provide a signal indicative of the pressure to the controller **430**.

The controller **430** is configured to use the signals from the pressure sensor **440** and the pressure sensor **442** to control the pump **100** throughout a cycle similar to that described herein with reference to FIGS. 11-16. The controller **430** is configured to position the DCV **420** in the first position, permitting fluid to pass through the passage A and into the inner chamber **122**. In response to the pressure sensed by the pressure sensor **440** reaching a threshold pressure (e.g., system pressure, etc.), the controller **430** is configured to move the DCV **420** to the second position, permitting fluid to pass through the passage B and into the inner chamber **222**. In response to the pressure sensed by the pressure sensor **442** exceeding the threshold pressure, the controller **430** is configured to move the DCV **420** back to the first position, causing the pump **100** to reciprocate.

In alternative embodiments, the pressure sensor **440** is configured to sense the pressure P_1 within the outer chamber **120** and/or the pressure sensor **442** is configured to sense the pressure P_3 within the outer chamber **220**. Due to the differing areas of each side of each piston, in such embodiments, the threshold pressures may be varied to achieve the

17

desired shifting characteristics. In another alternative embodiment, the pressure sensors **440**, **442** are omitted and a position sensor (e.g., a potentiometer, a linear variable differential transformer, one or more limit switches, etc.) is configured to provide a signal indicative of a position of the piston assembly **260**. In such an embodiment, the controller **430** may shift the position of the DCV **420** in response to signal from the position sensor indicating that the piston assembly **260** has reached a threshold X position (e.g., $X=-1$ inch or $X=+1$ inch, etc.).

Alternative Piston Geometries

Referring to FIG. **19**, the pump **100** is shown according to an alternative embodiment. In this embodiment, the piston **116** is replaced with a piston **500**. In the embodiment shown in FIGS. **5** and **7**, the piston **116** has a constant radius R_{P1} . In contrast, the piston **500** of FIG. **19** is stepped such that a first portion **502** of the piston **500** that is adjacent to the outer chamber **120** has a radius R_{P1O} . A second portion **504** of the piston **500** that is adjacent to the inner chamber **122** has a radius R_{P1I} . The first portion **502** has a larger radius than the second portion **504** and thus has a larger area. The piston tube **112** is sized (e.g., stepped) to create a seal with the piston **500**. In this embodiment, area A_{P1O} of the outer surface of the piston **500** and the area A_{P1I} of the inner surface of the piston can be calculated using the following equations:

$$A_{P1O} = \pi R_{P1O}^2 \quad (10)$$

$$A_{P1I} = \pi R_{P1I}^2 - \pi R_R^2 \quad (11)$$

By varying the sizes of each area, the force on the piston assembly **260** from the fluid in each chamber may be varied (e.g., as calculated using Equation 9). Accordingly, for a given system pressure, the output pressure of the pressurized fluid provided by the pump **100** may be varied. Similarly, for a given flow rate of fluid into the pump **100** from the pump controller **300**, the flow rate of fluid pressurized fluid provided by the pump **100** may be varied.

Referring to FIG. **20**, the pump **100** is shown according to another alternative embodiment. The pump **100** of FIG. **20** is substantially similar to the pump **100** of FIG. **19**, except the piston **500** is replaced with a piston **550**. The piston **550** is stepped such that a first portion **552** of the piston **550** that is adjacent to the outer chamber **120** has a radius R_{P1O} . A second portion **554** of the piston **550** that is adjacent to the inner chamber **122** has a radius R_{P1I} . The first portion **552** has a smaller radius than the second portion **554** and thus has a larger area. This arrangement facilitates customization of the pressures and flow rates from the pump **100**, similar to the arrangement of FIG. **19**. In this embodiment, the area of the inner surface of the piston **116** and the area of the outer surface of the piston **116** may be made equal (e.g., by increasing the size of the inner surface to account for the area of the piston rod **252**). Although the piston **500** and the piston **550** are shown as replacing the piston **116**, it should be understood that the piston **500** or the piston **550** may additionally or alternatively replace the piston **216**.

Inverted Chamber Arrangement

Referring to FIG. **21**, a pump **600** is shown according to an exemplary embodiment. The pump **600** may be substantially similar to the pump **100** except as specified herein. In the pump **600**, the functions of the inner and outer chambers are inverted. Specifically, the first fluid (e.g., hydraulic oil)

18

from the pump controller **300** is supplied to the outer chambers **120**, **220**, and the second fluid (e.g., nitrogen) is compressed within the inner chambers **122**, **222**. As shown, the passage A passes through the cap **110** and is fluidly coupled to the outer chamber **120**. The passage B passes through the cap **210** and is fluidly coupled to the outer chamber **220**. An inlet passage **602** and an outlet passage **604** extend through the main body **102** and are fluidly coupled to the inner chamber **122**. An inlet check valve **606** is positioned along the inlet passage **602**, and an outlet check valve **608** is positioned along the outlet passage **604**. An inlet passage **652** and an outlet passage **654** extend through the main body **102** and are fluidly coupled to the inner chamber **222**. An inlet check valve **656** is positioned along the inlet passage **652**, and an outlet check valve **658** is positioned along the outlet passage **654**. The inlet passage **602**, the outlet passage **604**, the inlet passage **652**, and the outlet passage **654** may function similarly to the inlet passage **130**, the outlet passage **132**, the inlet passage **230**, and the outlet passage **232**, respectively (e.g., by permitting nitrogen to enter and exit a corresponding chamber there-through). Accordingly, in this embodiment, the pump controller **300** may be fluidly coupled to the outer chamber **120** and the outer chamber **220** to control movement of the piston assembly **260**, and nitrogen may be compressed within the inner chamber **122** and the inner chamber **222**.

Series Pump

Referring to FIG. **22**, a pump **700** is shown according to an alternative embodiment. The pump **700** may be substantially similar to the pump **100** except as specified herein. In this embodiment, the outlet passage **132** is fluidly coupled to inlet passage **230** by a conduit (e.g., a hose, a pipe, a passage, etc.), shown as conduit **710**. Positioned along the length of the conduit **710** is a flow control device, shown as check valve **720**. The check valve **720** is configured to permit the flow of fluid from the outlet passage **132** to the inlet passage **230**. The check valve **720** is configured to prevent the flow of fluid from the inlet passage **230** to the outlet passage **132**. In other embodiments, one or more of the outlet check valve **136**, the inlet check valve **234**, and the check valve **720** are omitted.

The piston **216** is replaced with the piston **550** (e.g., as shown in FIG. **20**). This increases the pressure that the piston **550** exerts on the fluid within the outer chamber **220** relative to an embodiment that includes the piston **216**. Accordingly, the fluid leaving the outer chamber **220** can reach a higher pressure.

During operation, the piston assembly **260** moves in the positive X direction such that the piston **116** compresses a fluid (e.g., nitrogen, etc.) within the outer chamber **120**. As the pressure increases, some of the fluid is forced out of the outer chamber **120** through the outlet passage **132** and into the conduit **710**. The piston **216** expands the outer chamber **220**, drawing in fluid from the conduit **710**. When the movement of the piston assembly **260** changes direction, the piston **550** compresses the fluid within the outer chamber **220**. The inlet check valve **234**, the check valve **720**, and/or the outlet check valve **136** prevent the fluid from moving back into the outer chamber **120**. Low pressure fluid is drawn into the outer chamber **120** through the inlet passage **130**. Because the fluid passes in series from the outer chamber **120** to the outer chamber **220** during each cycle of the piston assembly **260**, the fluid experiences two cycles of compression prior to leaving the pump **700**. Due to this double compression and the stepped arrangement of the

piston **550**, the pump **700** can provide compressed fluid at a higher pressure than the pump **100**, but at a lower flow rate.

Pump Implementation in Vehicle Suspension System

In some embodiments, the pump system **302** is incorporated into a vehicle. Referring to the exemplary embodiment shown in FIGS. **23** and **24**, an axle assembly **1000** is configured to be included as part of a vehicle. The vehicle may be a military vehicle, a utility vehicle (e.g., a fire truck, a tractor, construction equipment, a sport utility vehicle, etc.), or still another type of vehicle. As shown in FIG. **23**, the axle assembly **1000** includes a differential **1012** coupled to a half shaft **1014**. As shown in FIG. **23**, the half shaft **1014** is coupled to a wheel-end assembly **1016**. The wheel-end assembly **1016** may include brakes, a gear reduction, steering components, a wheel hub, a wheel, a tire, and other features. According to an exemplary embodiment, the differential **1012** is configured to be coupled to a drive shaft of the vehicle. Such a differential **1012** may receive rotational energy from a prime mover (e.g., a diesel engine, a gasoline engine, an electric motor, etc.) of the vehicle. The differential **1012** then allocates torque provided by the prime mover between the half shafts **1014** of the axle assembly **1000**. The half shafts **1014** deliver the rotational energy to each wheel-end assembly **1016**. According to an alternative embodiment, each wheel-end assembly **1016** includes a prime mover (e.g., the axle assembly **1000** includes electric motors that each drive one wheel).

According to an exemplary embodiment, the axle assembly **1000** includes a suspension system **1018** that couples the chassis of the vehicle to wheel-end assembly **1016**. In some embodiments, the chassis includes a pair of opposing frame rails, and the suspension system **1018** engages the opposing frame rails through side plate assemblies. In other embodiments, the chassis is a hull, a capsule, or another type of structural member. According to an exemplary embodiment, the suspension system **1018** includes a spring, shown as gas spring **1020**, and a damper, shown as hydraulic damper **1022**. As shown in FIG. **23**, the gas spring **1020** and the hydraulic damper **1022** are coupled in parallel to a lower support member, shown as lower swing arm **1026**. According to an exemplary embodiment, the wheel-end assembly **1016** is coupled to lower swing arm **1026** and an upper support member, shown as upper swing arm **1024**.

According to an exemplary embodiment, the vehicle is configured for operation on both smooth (e.g., paved) and uneven (e.g., off-road, rough, etc.) terrain. As the vehicle travels over uneven terrain, the upper swing arm **1024** and the lower swing arm **1026** guide the vertical movement of the wheel-end assembly **1016**. A stop, shown as cushion **1028**, provides an upper bound to the movement of the wheel-end assembly **1016**. It should be understood that axle assembly **1000** may include similar components (e.g., wheel end assemblies, suspension assemblies, swing arms, etc.) for each of the two opposing lateral sides of a vehicle.

Referring next to the exemplary embodiment shown in FIG. **24**, the suspension system **1018** includes various components configured to improve performance of the vehicle. As shown in FIG. **24**, the gas spring **1020** is a high pressure gas spring. According to an exemplary embodiment, the suspension system **1018** includes the pump **100**, coupled to gas spring **1020**. Although the pump **100** is shown, it should be understood that any of the variants of the pump **100** (e.g., the pump **600**, the pump **700**, etc.) disclosed herein may be used instead of the pump **100**. In some embodiments,

suspension system **1018** includes a plurality of pumps **100** each coupled to a separate gas spring **1020**. In other embodiments, the suspension system **1018** includes fewer pumps **100** than gas springs **1020**. According to an exemplary embodiment, the gas spring and the pump include gas made up of at least 90% inert gas (e.g., nitrogen, argon, helium, etc.). The gas may be stored, provided, or received in one or more reservoirs (e.g., tank, accumulators, etc.). During operation, the pump **100** selectively provides gas, under pressure, to at least one of the gas springs **1020** and the reservoir. By providing gas to the gas springs **1020** or removing gas from the gas springs **1020**, the pump **100** can control a ride height of the vehicle and/or one or more response characteristics of the suspension system **1018** to an external stimulus (e.g., driving over a bump or hole, etc.).

Referring to FIGS. **25**, a hydraulic control assembly, shown as suspension control assembly **1100**, is shown according to an exemplary embodiment. The suspension control assembly **1100** is configured to control the flow of nitrogen to each gas spring **1020** of a vehicle. In this embodiment, the vehicle includes four suspension systems **1018**, one at each corner of the vehicle, and each including a gas spring **1020**. The suspension control assembly **1100** includes five ports: a port RR fluidly coupled to a right-rear gas spring **1020**; a port LR fluidly coupled to a left-rear gas spring **1020**; a port RF fluidly coupled to a right-front gas spring **1020**; a port LF fluidly coupled to a left-front gas spring **1020**; and a port TANK fluidly coupled to a low pressure gas source, shown as low pressure reservoir **1102**. Pressurized nitrogen passes into or out of each gas spring **1020** through the corresponding port, raising or lowering the ride height of the corresponding portion (e.g., corner) of the vehicle, respectively. The reservoir **1102** may be a tank, an accumulator, a pressure vessel, or any other type of container configured to contain and/or provide a relatively low pressure volume of nitrogen.

Referring to FIGS. **2** and **25**, a low pressure bus, shown as low pressure manifold **1110**, is fluidly coupled to the inlet passage **130** and the inlet passage **230** of the pump **100** to supply low pressure nitrogen to the outer chamber **120** and the outer chamber **220**. A high pressure bus, shown as high pressure manifold **1112**, is fluidly coupled to the outlet passage **132** and the outlet passage **232** of the pump **100** to receive high pressure nitrogen from the outer chamber **120** and the outer chamber **220** of the pump **100**. A series of flow control devices (e.g., check valves and directional control valves) control the flow of gas between (a) the low pressure manifold **1110** and the high pressure manifold **1112** and (b) each gas spring **1020** and the reservoir **1102**. As shown, the directional control valves are solenoid activated and spring returned. In other embodiments, the directional control valves are otherwise controlled. Actuation of the directional control valves and the pump may be controlled by an electronic suspension controller of the vehicle or by another type of controller.

To control the flow of gas to and from the port LF, the suspension control assembly **1100** includes a first directional control valve, shown as valve **S1A**, a second directional control valve, shown as valve **S1B**, a first check valve, shown as check valve **CV1**, and a second check valve, shown as check valve **CV2**. Each directional control valve is a two-position valve. Each directional control valve includes a check valve that (a) prevents flow of nitrogen therethrough in a first direction and (b) permits flow of nitrogen therethrough in a second direction opposite the first direction when the directional control valve is in a closed

21

position. When in an open position, each directional control valve permits flow therethrough in at least the first direction.

The check valve CV1 is configured to prevent flow of fluid from the port LF to the high pressure manifold 1112. Similarly, the check valve CV2 is configured to prevent flow of fluid from the low pressure manifold 1110 to the port LF. In the closed position, the valve S1A prevents flow of fluid from the high pressure manifold 1112 to the port LF. In the open position, the valve S1A fluidly couples the port LF to the high pressure manifold 1112. In the closed position, the valve S1B prevents flow of fluid from the port LF to the low pressure manifold 1110. In the open position, the valve S1B fluidly couples the port LF to the low pressure manifold 1110.

To add nitrogen to the gas spring 1020 of the port LF, the valve S1A can be opened to permit nitrogen from the high pressure manifold 1112 to enter into the port LF. To add nitrogen to the high pressure manifold 1112 (e.g., if the pressure in the high pressure manifold 1112 is too low, etc.), the pump 100 may be activated to introduce nitrogen into the high pressure manifold 1112. To remove nitrogen from the gas spring 1020 of the port LF, the valve S1B can be opened to permit nitrogen from the port LF to enter the low pressure manifold 1110. To remove nitrogen from the low pressure manifold 1110 (e.g., if the pressure in the low pressure manifold 1110 is too high, etc.), the pump 100 may be activated to draw nitrogen out of the low pressure manifold 1110.

Each other port of the suspension control assembly 1100 is coupled to similar valves to the port LF that can be similarly controlled to add nitrogen to the port or remove nitrogen from the port. Specifically, to perform similar functions to the check valve CV1, the check valve CV2, the valve S1A, and the S1B: the port RF is coupled to a check valve CV3, a check valve CV4, a valve S2A, and a valve S2B; the port LR is coupled to a check valve CV5, a check valve CV6, a valve S3A, and a valve S3B; the port RR is coupled to a check valve CV7, a check valve CV8, a valve S4A, and a valve S4B; and the port TANK is coupled to a check valve CV9, a check valve CV0, a valve STA, and a valve STB. By opening closing these valves, nitrogen may be moved between the gas springs 1020, the reservoir 1102, and the pump 100. In some embodiments, the assembly of the suspension control assembly 1100, the pump 100, the gas springs 1020, and the reservoir 1102 is a closed system containing a fixed amount of nitrogen.

Referring to FIGS. 2 and 25, in this arrangement, the pump 100 is always coupled to the low pressure manifold 1110 and the high pressure manifold 1112. Due to the coupling arrangement provided by the inlet check valve 134 and the inlet check valve 234, the pressure within the outer chamber 120 and the outer chamber 220 is always at least the pressure of the low pressure manifold 1110. During operation, when one outer chamber becomes pressurized, the other maintains the pressure of the low pressure manifold 1110. As shown in Equation 9, this pressure forces the piston assembly 260 toward the other outer chamber. This increases the compressive force on this outer chamber, which in turn increases the maximum output pressure of the pump 100.

Beneficially, the delay introduced by the orifice 334 and the orifice 344 and the minimum pressures within the outer chamber 120 and the outer chamber 220 cooperate to increase the maximum output pressure of the pump 100. In one embodiment having a similar structure to that of FIG. 11, the area A_{P1O} and the area A_{P2O} are both equal to 3.14 in², and the area A_{P1I} and the area A_{P2I} are both equal to 2.7

22

in². The outer chamber 120 and outer chamber 220 both have a minimum pressure of 450 psi. The system pressure is 3250 psi, and the shift pressure is 3100 psi.

In this embodiment, the additional force contributed by the minimum pressure is approximately 1413 lbf. This additional force is enough to make up for the reduction in force caused by the area of the piston rod 252. In this embodiment, the force imparted on the piston assembly 260 by fluid within one of the inner chambers at the shift pressure is approximately 8370 lbf, whereas the force imparted on the piston assembly 260 by fluid within one of the inner chambers at system pressure is approximately 8774 lbf. Accordingly, by using orifices to introduce a shifting delay, an additional 404 lbf is gained without having to increase the size of the pump 100.

As utilized herein, the terms “approximately,” “about,” “substantially,” and similar terms are intended to have a broad meaning in harmony with the common and accepted usage by those of ordinary skill in the art to which the subject matter of this disclosure pertains. It should be understood by those of skill in the art who review this disclosure that these terms are intended to allow a description of certain features described and claimed without restricting the scope of these features to the precise numerical ranges provided. Accordingly, these terms should be interpreted as indicating that insubstantial or inconsequential modifications or alterations of the subject matter described and claimed are considered to be within the scope of the disclosure as recited in the appended claims.

It should be noted that the term “exemplary” and variations thereof, as used herein to describe various embodiments, are intended to indicate that such embodiments are possible examples, representations, or illustrations of possible embodiments (and such terms are not intended to connote that such embodiments are necessarily extraordinary or superlative examples).

The term “coupled” and variations thereof, as used herein, means the joining of two members directly or indirectly to one another. Such joining may be stationary (e.g., permanent or fixed) or moveable (e.g., removable or releasable). Such joining may be achieved with the two members coupled directly to each other, with the two members coupled to each other using a separate intervening member and any additional intermediate members coupled with one another, or with the two members coupled to each other using an intervening member that is integrally formed as a single unitary body with one of the two members. If “coupled” or variations thereof are modified by an additional term (e.g., directly coupled), the generic definition of “coupled” provided above is modified by the plain language meaning of the additional term (e.g., “directly coupled” means the joining of two members without any separate intervening member), resulting in a narrower definition than the generic definition of “coupled” provided above. Such coupling may be mechanical, electrical, or fluidic.

References herein to the positions of elements (e.g., “top,” “bottom,” “above,” “below”) are merely used to describe the orientation of various elements in the FIGURES. It should be noted that the orientation of various elements may differ according to other exemplary embodiments, and that such variations are intended to be encompassed by the present disclosure.

The present disclosure contemplates methods, systems and program products on any machine-readable media for accomplishing various operations. The embodiments of the present disclosure may be implemented using existing computer processors, or by a special purpose computer processor

23

for an appropriate system, incorporated for this or another purpose, or by a hardwired system. Embodiments within the scope of the present disclosure include program products comprising machine-readable media for carrying or having machine-executable instructions or data structures stored thereon. Such machine-readable media can be any available media that can be accessed by a general purpose or special purpose computer or other machine with a processor. By way of example, such machine-readable media can comprise RAM, ROM, EPROM, EEPROM, or other optical disk storage, magnetic disk storage or other magnetic storage devices, or any other medium which can be used to carry or store desired program code in the form of machine-executable instructions or data structures and which can be accessed by a general purpose or special purpose computer or other machine with a processor. Combinations of the above are also included within the scope of machine-readable media. Machine-executable instructions include, for example, instructions and data which cause a general purpose computer, special purpose computer, or special purpose processing machines to perform a certain function or group of functions.

Although the figures and description may illustrate a specific order of method steps, the order of such steps may differ from what is depicted and described, unless specified differently above. Also, two or more steps may be performed concurrently or with partial concurrence, unless specified differently above. Such variation may depend, for example, on the software and hardware systems chosen and on designer choice. All such variations are within the scope of the disclosure. Likewise, software implementations of the described methods could be accomplished with standard programming techniques with rule-based logic and other logic to accomplish the various connection steps, processing steps, comparison steps, and decision steps.

It is important to note that the construction and arrangement of the pump system as shown in the various exemplary embodiments is illustrative only. Additionally, any element disclosed in one embodiment may be incorporated or utilized with any other embodiment disclosed herein. For example, the piston 500 of the exemplary embodiment shown in at least FIG. 19 may be incorporated in the pump 600 of the exemplary embodiment shown in at least FIG. 21. Although only one example of an element from one embodiment that can be incorporated or utilized in another embodiment has been described above, it should be appreciated that other elements of the various embodiments may be incorporated or utilized with any of the other embodiments disclosed herein.

What is claimed is:

1. A pump system comprising:

a housing defining a first internal volume and a second internal volume;

a first piston positioned to separate the first internal volume into a first chamber and a second chamber;

a second piston positioned to separate the second internal volume into a third chamber and a fourth chamber;

a first inlet check valve configured to be at least fluidly coupled to a gas spring, the first inlet check valve positioned to permit a first fluid received from the gas spring to enter the first chamber;

a second inlet check valve configured to be at least fluidly coupled to the gas spring, the first inlet check valve positioned to permit the first fluid received from the gas spring to enter the third chamber;

a directional control valve (DCV) repositionable between (a) a first position where the DCV is configured to

24

fluidly couple the second chamber to a second fluid from a high pressure fluid source and (b) a second position where the DCV is configured to fluidly couple the fourth chamber to the second fluid from the high pressure fluid source; and

a relief valve configured to supply the second fluid to the DCV through an orifice to move the DCV from the first position to the second position, wherein the orifice is separate from the relief valve and positioned to resist flow of the second fluid to the relief valve.

2. The pump system of claim 1, wherein the second fluid is supplied by the high pressure fluid source.

3. The pump system of claim 1, wherein the relief valve is fluidly coupled to the second chamber through the orifice.

4. The pump system of claim 1, further comprising a piston rod extending between the first piston and the second piston such that a movement of the first piston causes an equal movement of the second piston.

5. The pump system of claim 4, wherein the housing includes a divider separating the second chamber from the fourth chamber, and wherein the divider defines a piston rod aperture through which the piston rod extends.

6. The pump system of claim 1, wherein the first inlet check valve is directly fluidly coupled to the first chamber throughout an entire range of motion of the first piston, and wherein the second inlet check valve is directly fluidly coupled to the third chamber throughout the entire range of motion of the second piston.

7. The pump system of claim 1, wherein the housing defines a first outlet passage fluidly coupled to the first chamber and a second outlet passage fluidly coupled to the third chamber, and wherein the first outlet passage is fluidly coupled to the second outlet passage.

8. The pump system of claim 1, wherein the first chamber contains a gas, and wherein the second chamber contains a liquid.

9. The pump system of claim 1, wherein the first piston and the second piston are configured to translate along an actuation axis, and wherein the first piston and the second piston are centered about the actuation axis.

10. The pump system of claim 1, wherein the second chamber and the fourth chamber are positioned between the first chamber and the third chamber.

11. A pump system comprising:

a housing defining a first internal volume and a second internal volume;

a first piston positioned to separate the first internal volume into a first chamber and a second chamber;

a second piston positioned to separate the second internal volume into a third chamber and a fourth chamber, wherein the first chamber and the third chamber are configured to receive a first fluid from a gas spring at a low pressure, and the second chamber and the fourth chamber are configured to receive a second fluid from a second fluid source;

a directional control valve (DCV) repositionable between (a) a first position where the DCV is configured to fluidly couple the second chamber to a reservoir and (b) a second position where the DCV is configured to fluidly couple the fourth chamber to the reservoir; and a relief valve configured to supply the second fluid through an orifice to move the DCV between the first position and the second position in response to a pressure at a point between the relief valve and the orifice reaching a preset shift pressure.

12. The pump system of claim 11, wherein the DCV is configured to fluidly couple a fluid source to the fourth

25

chamber in the first position, wherein the DCV is configured to fluidly couple the fluid source to the second chamber in the second position, and wherein the fluid is supplied by the fluid source, and wherein when the second fluid is supplied to the second chamber or the fourth chamber, the first fluid is forced out of one of the first chamber or the third chamber at a high pressure by one of the first piston or the second piston and into the gas spring.

13. The pump system of claim 11, wherein the relief valve is fluidly coupled to the second chamber through the orifice.

14. The pump system of claim 11, wherein the relief valve is a first relief valve, further comprising a second relief valve fluidly configured to supply the second fluid to the DCV to move the DCV between the second position and the first position.

15. A pump system comprising:

a housing defining a first internal volume and a second internal volume;

a first piston positioned to separate the first internal volume into a first chamber and a second chamber;

a second piston positioned to separate the second internal volume into a third chamber and a fourth chamber, wherein the first chamber and the third chamber are configured to receive a first fluid from a gas spring at a low pressure, and the second chamber and the fourth chamber are configured to receive a second fluid from a second fluid source;

a directional control valve (DCV) fluidly coupled to the second chamber and the fourth chamber;

a relief valve fluidly coupled to the DCV via a control line and the second chamber via a sensing line; and

an orifice positioned along the sensing line, wherein the relief valve is configured to reposition the DCV in response to a pressure at a point between the relief valve and the orifice reaching a shift pressure, and wherein the orifice is positioned to cause a delay in a flow of a fluid along the sensing line, such that a

26

pressure within the second chamber exceeds the shift pressure when the pressure at the point between the relief valve and the orifice reaches the shift pressure.

16. The pump system of claim 15, wherein the DCV is repositionable between (a) a first position where the DCV fluidly couples the second chamber to a reservoir and (b) a second position where the DCV fluidly couples the fourth chamber to the reservoir.

17. The pump system of claim 15, wherein the DCV is repositionable between (a) a first position where the DCV fluidly couples the fourth chamber to the second fluid source and (b) a second position where the DCV fluidly couples the second chamber to the second fluid source.

18. The pump system of claim 1, wherein the relief valve is configured to supply the second fluid to the DCV through the orifice to move the DCV from the first position to the second position in response to a pressure at a point between the relief valve and the orifice exceeding a preset shift pressure.

19. The pump system of claim 1, wherein the relief valve is configured to supply the second fluid to the DCV through the orifice to move the DCV from the first position to the second position in response to a pressure at a point between the relief valve and the orifice reaching a shift pressure, wherein the high pressure fluid source is configured to supply the second fluid at a system pressure that is higher than the shift pressure, and wherein the orifice is positioned such that a pressure within the second chamber reaches the system pressure before the pressure at the point between the relief valve and the orifice reaches the shift pressure.

20. The pump system of claim 18, wherein the pressure within the second chamber exceeds the shift pressure by a preset pressure difference when the pressure at the point between the relief valve and the orifice reaches the shift pressure.

* * * * *